

Effects of nicotine compared to placebo gum on sensitivity to pain and mediating effects of peak alpha frequency

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Samantha K Millard , Alan KI Chiang, Peter Humburg, Nahian Chowdhury, Raafay Rehan, Andrew J Furman, Ali Mazaheri, Siobhan M Schabrun, David A Seminowicz

Faculty of Medicine, University of New South Wales (UNSW), Sydney, Australia • Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia • Mark Wainwright Analytical Centre, University of New South Wales (UNSW), Sydney, Australia • Department of Radiology and Nuclear Medicine, University of Maryland School of Medicine, Baltimore, United States • Department of Neural and Pain Sciences, School of Dentistry, University of Maryland Baltimore, Baltimore, United States • Center to Advance Chronic Pain Research, University of Maryland Baltimore, Baltimore, United States • School of Psychology, University of Birmingham, Birmingham, United Kingdom • Centre for Human Brain Health (CHBH), University of Birmingham, Birmingham, United Kingdom • The Gray Centre for Mobility and Activity, Parkwood Institute, London, Canada • School of Physical Therapy, University of Western Ontario, London, Canada • Department of Medical Biophysics, Schulich School of Medicine & Dentistry, University of Western Ontario, London, Canada

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eLife Assessment

In this **useful** study, Millard et al. assessed the effects of nicotine on pain sensitivity and peak alpha frequency (PAF). The evidence shown is **incomplete** to support the key claim that nicotine modulates PAF or pain sensitivity, considering the effect sizes observed. This raises the question of whether the chosen experimental intervention was the most suitable approach for investigating their research question. Nonetheless, the work can be incorporated into the literature investigating the relationship between nicotine and pain, and could be of broad interest to pain researchers.

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Abstract

Recent research has linked individual peak alpha frequency (PAF) to pain sensitivity, but whether PAF alterations can influence pain remains unclear. Our study investigated the effects of nicotine on pain sensitivity and whether pain changes are mediated by PAF changes. In a randomised, double-blind, placebo-controlled experiment, 62 healthy adults (18–44 years) received either 4 mg nicotine gum (n=29) or placebo (n=33). Resting state EEG and pain ratings during prolonged heat and pressure models were collected before and after nicotine intake. The nicotine group had a decrease in heat pain ratings compared to the placebo group and increased PAF speed across the scalp from pre to post-gum, driven by

changes at central-parietal and right-frontal regions. However, mediation analysis did not support the notion that PAF changes mediate nicotine's effects on pain sensitivity. While a growing body of literature supports a link between PAF and both acute and chronic pain, further work is needed to understand the mechanisms of this link.

1 Introduction

Chronic pain is a highly prevalent and multifaceted health issue, which produces significant physical, emotional, and economic burden to those in pain, their families and friends, health care systems, and society [1–3]. Identifying modifiable factors that influence pain sensitivity could be a key step in reducing the presence and burden of chronic pain [4–6]. Individual peak alpha frequency (PAF) is an electro-physiological brain measure that shows promise as a biomarker of pain sensitivity [7–13]. Although the number of studies investigating PAF as a predictor of pain is increasing [14–18], no experiments have manipulated PAF while also measuring pain, which has limited our understanding of whether, and how, PAF is mechanistically linked to pain sensitivity.

PAF refers to the dominant oscillatory frequency in the 8–12 Hz alpha range. As alpha oscillations are believed to represent periodic inhibitory processes that direct temporal aspects of sensory information processing [19,20], it is thought that PAF reflects the rate at which sensory information is sampled from the external environment [21–23]. In the context of pain, slower PAF speed has been associated with the presence and duration of chronic pain [17,24–28], while in healthy, pain-free individuals, slower PAF has been associated with higher pain sensitivity in experimental pain models [8–10] and acute post-surgical settings [11].

While previous studies suggest that PAF could be used as either a biomarker for disease progression or as a predictive biomarker for pain sensitivity, critical gaps remain in our understanding of how PAF is mechanistically linked to pain. Specifically, no study has examined whether the manipulation of PAF alters pain sensitivity. Addressing this question has significant implications, as it would determine i) whether PAF is causally implicated in pain processing, and ii) whether manipulation of PAF has relevance as a method of clinical pain modulation. Several methods have been used to alter individual PAF, including brain stimulation [29–31], exercise [32–34], visual stimulation [35,36], cognitive tasks [37], and nicotine [38–40]. Because evidence suggests that nicotine can modulate PAF, where both nicotine and smoking increase PAF speed [38,41–48], we chose nicotine to assess our aim of whether change in PAF mediate changes in pain in a ‘mediation by design’ approach [49]. In addition, given evidence that nicotine

may increase experimental pain thresholds and tolerance [50–54], nicotine could also influence pain ratings during tonic pain. Thus, the primary questions addressed in this study were whether nicotine can: i) increase peak alpha frequency (PAF), ii) reduce pain sensitivity, and iii) determine whether changes in pain sensitivity are mediated by changes in PAF speed.

A secondary question pertains to the nature of the PAF–pain relationship. A substantial proportion of the current literature suggests that slower PAF speed is associated with higher pain sensitivity [8–11,17,24–27]. However, conflicting findings exist, with some studies reporting no relationship between PAF and chronic pain [55–58], or an opposite relationship where faster PAF is associated with higher pain sensitivity in experimental pain settings [7,15]. Divergent results could be due to variations in the relationship between PAF and different types of pain, or the dependence of this relationship on individual factors. For example, it was recently suggested that PAF–pain relationships may only be observed in neuropathic pain or expressed differently in males and females [17]. One way to address this issue is to investigate PAF across multiple pain models [12]; currently the PAF–pain relationship has been studied using thermal

[7,8,10], musculoskeletal [9,15], and surgical [11] pain, but never with multiple pain modalities included in the same study. Thus, a secondary question was whether baseline PAF, measured in a pain-free resting state, correlated with pain sensitivity during a model of heat pain (i.e. PHP), which has been previously assessed and is negatively correlated with PAF [10], and a model of pressure pain (i.e. cuff pressure algometry; CPA [59–63]) that has not been assessed in relation to PAF. Applying two experimental pain models also allows assessment of the generalisability of nicotine's effects on pain and the potential mediating effects of PAF, including whether these effects are consistent across different types of pain.

Using a randomised, placebo-controlled, double-blind, parallel design, a sample of healthy, pain-free, non-smoking participants had their resting state PAF measured before and after two models of prolonged pain (i.e. CPA and PHP; pre-gum). Participants then chewed either nicotine or placebo gum, after which their resting state PAF was re-assessed before and after the two prolonged pain models (i.e. CPA and PHP; post-gum). We predicted that: 1. nicotine gum would increase PAF speed in

comparison to placebo gum; 2. nicotine gum would decrease pain ratings in comparison to placebo gum; 3. decreases in pain ratings following nicotine gum would be mediated by increases in the speed of PAF; and 4. faster baseline PAF would be associated with lower baseline pain ratings. Based on a pre-registered analysis plan (<https://osf.io/pc4rq/>), predictions 1–3 were tested using two-wave latent change score (2W-LCS) mediation models [64–66], while prediction 4 was tested using correlation analysis.

2 Results

The statistical analysis plan for this study was pre-registered on Open Science Framework (OSF: <https://osf.io/pc4rq/>) in July 2022.

2.1 Sample Characteristics

The participant sample included 62 healthy adults (26.45 ± 7.28 ; range: 18–44 years), without neurological or psychological disorders. Fourteen participants self-identified as North-West European (22.58%), 12 as South-East Asian, (19.36%), and eight reported mixed ancestries (12.90%; see Supplementary Material 1 for a full list of self-reported ancestry). There were 32 females (24.53 ± 6.52 ; range: 18–43 years) and 30 males (28.5 ± 7.58 ; range: 19–44 years); no participants reported their sex at birth as inter-sex. Thirty-one individuals identified as women, 30 as men, and 1 as non-binary.

Regarding pain in the past two weeks, 20 participants reported “None” (nicotine: $n = 10$; placebo: $n = 10$), 16 reported “Very Mild” (nicotine: $n = 14$; placebo: $n = 12$), 15 reported “Mild” (nicotine: $n = 5$; placebo: $n = 10$), one reported “Moderate” (nicotine: $n = 1$, placebo: $n = 0$), and no participants reported “Severe” pain in the past two weeks. Participants were successfully blinded to group allocation (Supplementary Material 2) and the remaining baseline demographics are displayed in **Table 1**. Following CONSORT recommendations [67,68], no baseline statistics are reported.

2.2 Effects of nicotine gum on alpha oscillations and prolonged pain ratings

Mediation analysis was conducted to assess the effects of nicotine on PAF and pain ratings, and whether change in PAF mediated change in pain.

Group	N = 62	Nicotine (n = 29)	Placebo (n = 33)
Age	26.45 (± 7.28)	26.18 (± 8.06)	26.39 (± 6.65)
Weight (kg)	68.16 (± 13.42)	68.9 (± 13.00)	67.5 (± 13.90)
Height (cm)	172.29 (± 10.08)	172.40 (± 10.00)	172.20 (± 10.30)
Depressive	0.53 (± 0.97)	0.62 (± 1.10)	0.46 (± 0.75)
Anxiety	0.69 (± 1.06)	0.76 (± 1.30)	0.64 (± 0.82)
Stress	9.82 (± 3.81)	8.93 (± 4.04)	10.61 (± 3.48)
Sleep quality	6.97 (± 1.77)	6.79 (± 1.86)	7.12 (± 1.71)
Heat pain threshold	41.27 (± 2.59)	40.97 (± 2.79)	41.54 (± 2.41)
Pressure threshold	293.92 (± 121.21)	318.70 (± 111.25)	272.14 (± 127.00)
Alpha power (8–12 Hz)	26.24 (± 7.02)	25.89 (± 7.64)	26.55 (± 6.54)

Note. PAF = peak alpha frequency; PHP = phasic heat pain; CPA = cuff pressure algometry.

Table 1

Mean $\pm$ standard deviations (SD) for baseline participant characteristics split by gum group.

ANCOVA-equivalent two-wave latent change score (2W-LCS) models (i.e. with autoregressive and coupling parameters) and difference score 2W-LCS models (i.e. without auto-regressive and coupling parameters) were constructed for mean PHP and CPA pain outcomes separately (see Methods for further details). As no significant difference was found between model fits (see Supplementary Materials 3.1 and 3.2), the difference score 2W-LCS models are reported here; whilst the full ANCOVA-equivalent 2W-LCS models are presented in Supplementary Material 3.1.

Nicotine gum modulated global PAF (8–12 Hz) and PHP ratings, but change in PHP ratings were not mediated by change in PAF

The PHP model consisted of five repetitions of a 40 second heat stimulation to the inside left forearm at a fixed 46 °C (see Methods), which was most frequently described as moderate/severe hot, sharp, and shooting pain (**Figure 1**, plot A). Raw time courses of pain ratings during PHP are displayed in **Figure 1** (plot C), from which the mean pain during max heat stimulation was calculated (**Table 2**, **Figure 1**, plot E).

Five participants could not tolerate the fixed 46 °C stimulus during familiarisation, therefore a lower temperature of 45 °C was used for both pre- and post-gum PHP assessments for these participants, as reported in the protocol (see Methods). Two participants were excluded from mediation analysis assessing the effects of nicotine on PHP ratings: one from the placebo group due to an equipment issue during the second pain assessment, and one from the nicotine group that could not tolerate the baseline temperature for the second pain assessment (nicotine: $n = 28$; placebo: $n = 32$).

The difference score model for PHP (**Figure 3**; model fit: $\chi^2(24) = 25.874$, $p = 0.36$, CFI = 0.99, RMSEA = 0.036, SRMR = 0.078) suggests a significant total ($b = -0.62$, $p = .023$, bootstrapped 95% CI: [-1.21, -0.15]) and direct ($b = -0.68$, $p = .021$, bootstrapped 95% CI: [-1.28, -0.16]) effect of nicotine on change in PHP ratings (**Figure 1**). Change in pain was reduced by a factor of -0.667 for the nicotine group compared to the placebo group when controlling for confounding variables (**Figure 3**).

Frequency spectra (plots A and B) and topographical plots (plot E) of PAF pre-post nicotine, are displayed by group in **Figure 2**, with PAF values displayed in **Table 3**. The difference score model suggests a significant effect of nicotine gum on change in global PAF ($b = 0.085$, $p = .018$, bootstrapped 95% CI: [0.017, 0.15]), suggesting change in PAF was increased by a factor of 0.085 for the nicotine group compared to placebo when controlling for confounding variables (**Figure 3**). Therefore, nicotine changed PAF and PHP ratings in the predicted directions compared to what would be expected without nicotine (i.e. placebo). However, as the estimated effect of change in PAF speed on change in PHP ratings was non-significant ($b = 0.23$, $p = .78$, bootstrapped 95% CI: [-1.28, 1.9]), the change in PAF did not mediate the change in PHP ratings (i.e. indirect effect; $b = 0.019$, $p = .81$, bootstrapped 95% CI: [-0.093, 0.23]).

There was also a significant effect of perceived stress at baseline on change in PHP ratings when controlling for group allocation and other confounding variables ($b = -0.096$, $p = .048$, bootstrapped 95% CI: [-0.19, -0.000047]), where higher perceived stress resulted in larger decreases in PHP ratings (see Supplementary Material 3.3 for post-hoc analysis of stress). There was also a significant effect of sex on change in PAF ($b = -0.077$, $p = .037$, bootstrapped 95% CI: [-0.16, -0.011]); change in PAF was reduced by a factor of -0.077 for males compared to females when controlling for gum group and confounders, therefore the increase in PAF speed was greater for females than males (see Supplementary Material 3.4 for post-hoc analysis of sex differences).

A complete table of PAF and power values for all four time points is displayed in Supplementary Material 4.1. Exploratory analysis of interactions between time point (i.e. pre-/post-gum) and group (i.e. nicotine/placebo) on PHP ratings, PAF speed, PHP ratings, and alpha power, using

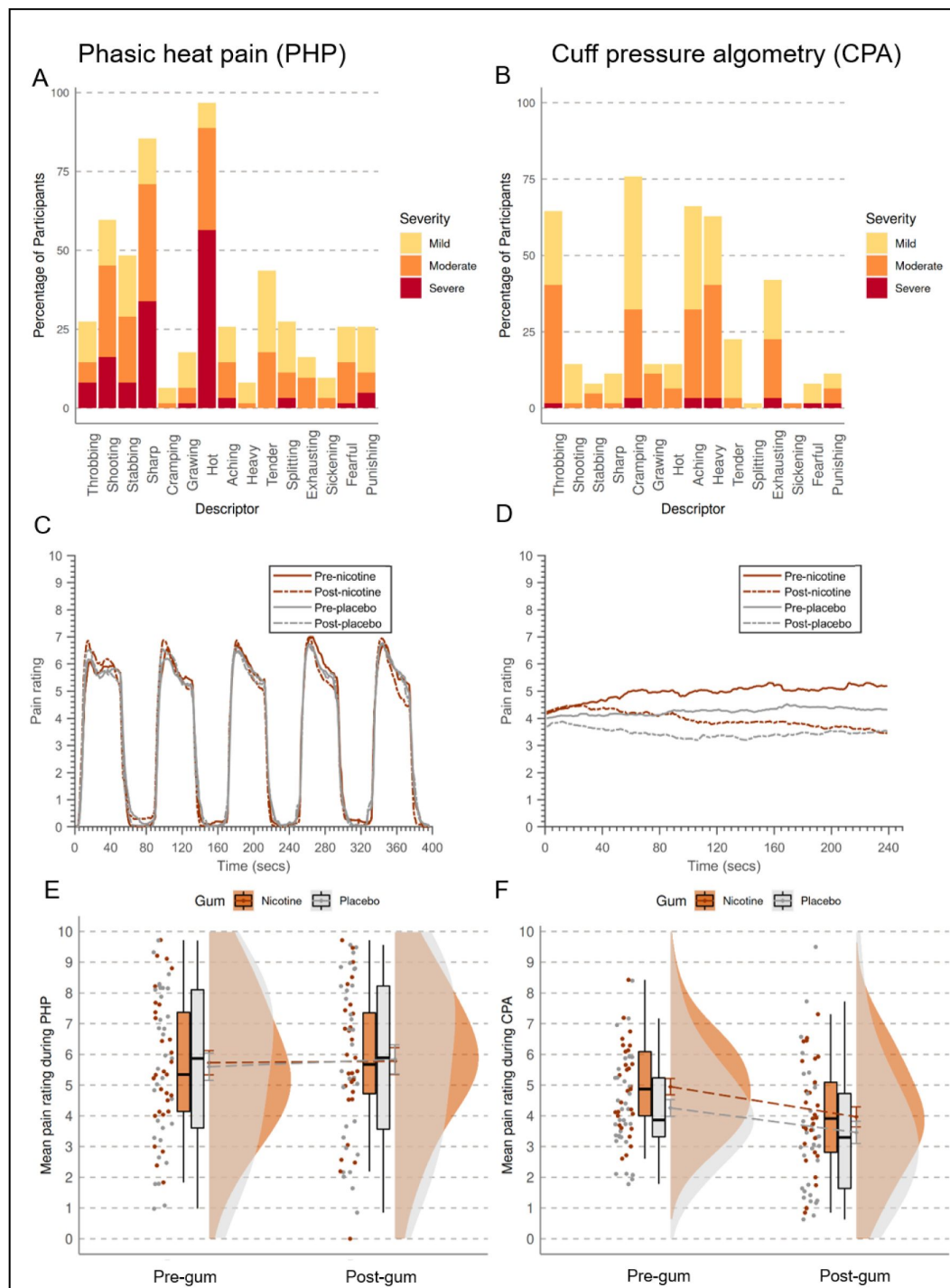


Figure 1

Percentage of participants reporting each category descriptor of pain on the McGill pain questionnaire for A) phasic heat pain (PHP) and B) cuff pressure algometry (CPA) at baseline. Average pain time courses split into nicotine (orange, $n = 29$) and placebo (grey, $n = 33$) gum groups for C) PHP and D) CPA models of prolonged pain. Raincloud plots of mean pain ratings during E) PHP and F) CPA models of prolonged pain.

Gum group	n	PHP		n	CPA	
		Pre-gum	Post-gum		Pre-gum	Post-gum
Total	62	5.66 (± 2.34)		62	4.58 (± 1.53)	3.70 (± 1.93)
	60	5.66 (± 2.32)	5.81 (± 2.50)			
Nicotine	29	5.73 (± 2.13)		29	4.95 (± 1.43)	3.97 (± 1.76)
	28	5.83 (± 2.10)	5.78 (± 2.33)			
Placebo	33	5.84 (± 2.67)		33	4.26 (± 1.57)	3.46 (± 2.07)
	32	5.51 (± 2.53)	5.84 (± 2.67)			

Table 2

Mean $\pm$ standard deviations (SD) for pain ratings during phasic heat pain (PHP) and cuff pressure algometry (CPA), for pre and post-gum, for the whole sample and separated by nicotine and placebo groups. Note that post-gum PHP ratings could not be collected for two participants. Therefore, the full sample of 62 participants were used for baseline correlation analyses of PHP and CPA, as well as for CPA mediation models, but list-wise deletion to ensure full pre-post samples meant that only 60 participants were used for PHP mediation models.

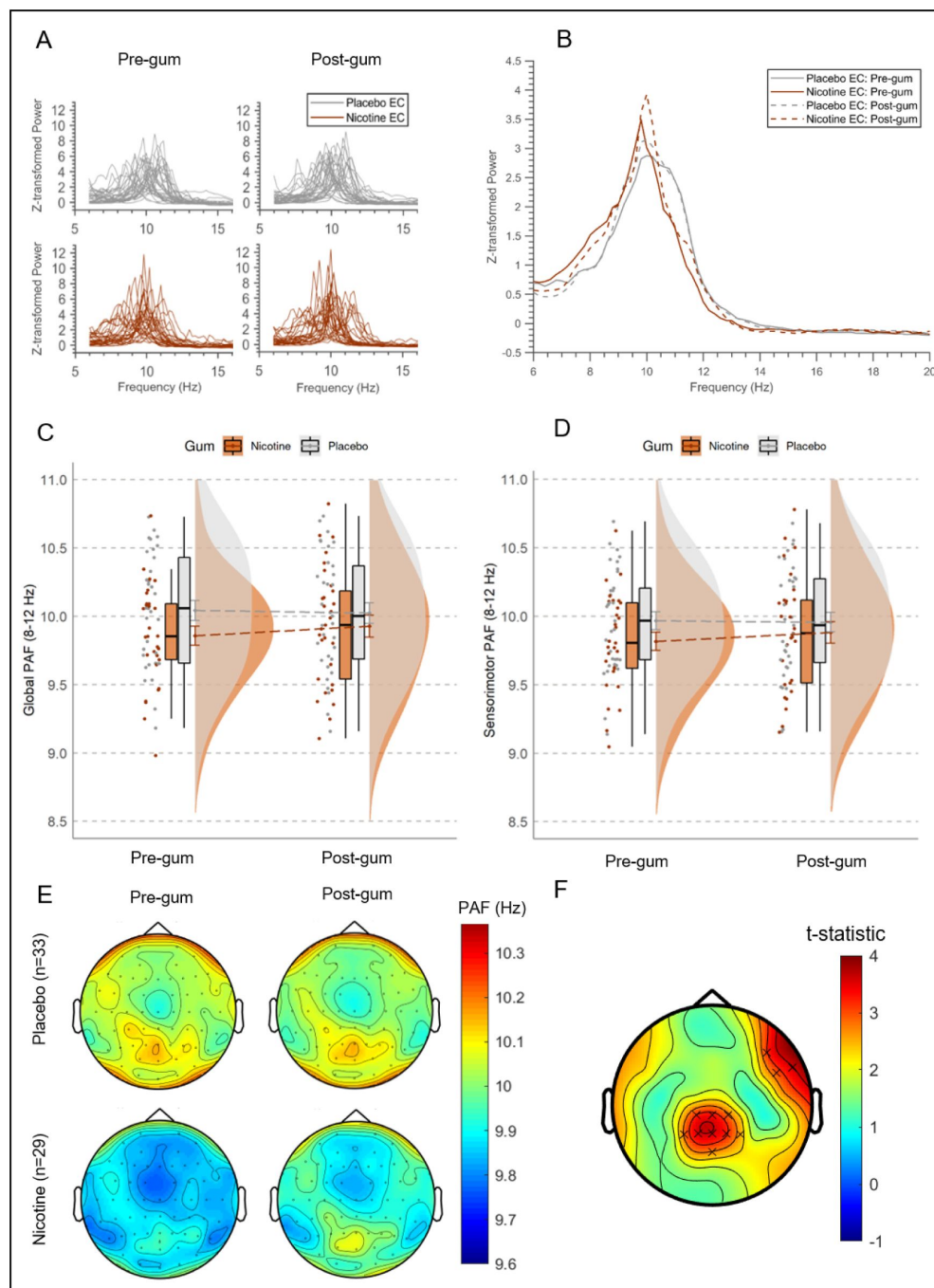


Figure 2

A) Individual spectral plots of eyes closed (EC) resting state EEG data showing dominant peaks within the alpha range of 8–12 Hz. Note that many participants' peaks fall outside the narrow 9–11 Hz window. Top panel shows individual spectra for participants in the placebo group (in grey, $n = 33$), bottom panel shows individual spectra for participants in the nicotine group (in orange, $n = 29$). B) Spectral plots of eyes closed (EC) resting state EEG shows averaged spectra by group and time point. C) Raincloud plots of mean global and D) sensorimotor PAF values, calculated with a wide window (8–12 Hz) and the centre of gravity method, before and after chewing either nicotine (in orange) or placebo (in grey) gum. E) Topographical plots of PAF calculated for each electrode with the centre of gravity method and 8–12 Hz window are shown for the pre-gum baseline (rest 1) and post-gum (rest 3). F) Topographical plot of t-test statistics produced by cluster-based permutation analysis of nicotine group, displaying significant difference in PAF after chewing nicotine gum in two separate clusters marked by crosses (x).

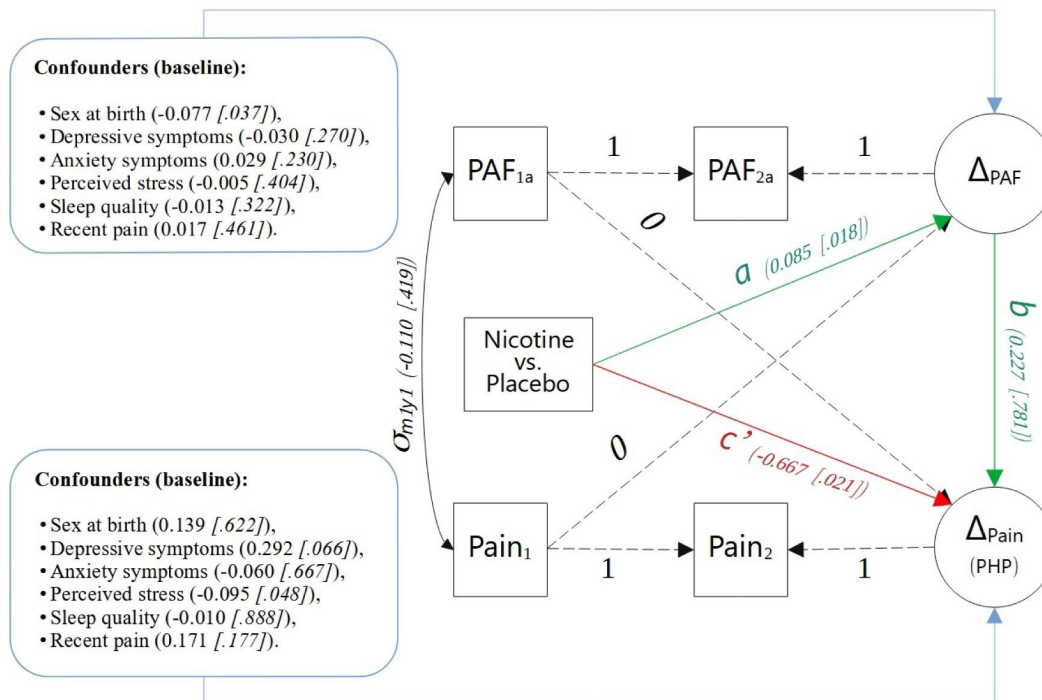


Figure 3

Path model displaying unstandardised estimates and p-values for the difference score two-wave latent change score (2W-LCS) model assessing the effect of nicotine against placebo (i.e. gum group) on change in global 8–12 Hz peak alpha frequency (i.e. Δ_{PAF}) and change in mean phasic heat pain (PHP) ratings (i.e. Δ_{Pain}). Green arrows highlight the indirect effects (i.e. a and b), the red arrow highlights the direct effect (i.e. c'), the blue lines highlight pre-determined confounding variables, and dashed lines denote fixed effects. This model suggests that chewing nicotine gum, compared to placebo, increases the speed of PAF, and decreases PHP ratings, however the change in PAF does not influence the change in PHP ratings, therefore changes in PHP ratings after nicotine gum are not mediated by changes in PAF.

Gum group	Sensorimotor					
	PAF			Global PAF		
	n	Pre-gum	Post-gum	n	Pre-gum	Post-gum
Total	62	9.90 (± 0.37)	9.92 (± 0.41)	62	9.96 (± 0.41)	9.98 (± 0.43)
	60	9.90 (± 0.38)	9.93 (± 0.41)	60	9.96 (± 0.41)	9.98 (± 0.44)
Nicotine	29	9.82 (± 0.36)	9.88 (± 0.42)	29	9.86 (± 0.37)	9.93 (± 0.44)
	28	9.81 (± 0.36)	9.87 (± 0.43)	28	9.85 (± 0.38)	9.92 (± 0.44)
Placebo	33	9.97 (± 0.38)	9.96 (± 0.41)	33	10.04 (± 0.42)	10.02 (± 0.43)
	32	9.98 (± 0.38)	9.97 (± 0.40)	32	10.05 (± 0.42)	10.04 (± 0.43)

Table 3

Mean $\pm$ standard deviations (SD) for sensorimotor and global peak alpha frequency (PAF) pre- and post-gum chewing, for the whole sample and by nicotine and placebo groups. PAF calculated for wide frequency window (8–12 Hz) with the centre of gravity method. Note that the 62 participants is the full sample that was used for mediation analysis of cuff pressure algometry (CPA) pain ratings and baseline correlations of PAF with CPA and phasic heat pain (PHP), whilst the 60 participants is the sample used for mediation analysis of PHP, as two participants were deleted list wise due to missing post-gum PHP assessments, one from each gum group.

repeated measures analysis of variance (RM-ANOVA) that do not control for confounding, are displayed in Supplementary Materials 4 and Notably, exploratory RM-ANOVAs still indicated interactions between gum group and time point on PAF (8–12 Hz) indicating that PAF only increased for the nicotine group after chewing (Supplementary Material 4), but there was no interaction between gum group and time point on PHP ratings (Supplementary Material 5), underscoring the necessity of controlling for confounding variables.

Nicotine gum modulated global PAF (8–12 Hz), with no total, direct, or indirect effects on CPA ratings

The CPA model involved four minutes of continuous cuff pressure to the left calf muscle that was initially calibrated to each individual's 4/10 pain (see Methods), and was most frequently described as mild/moderate cramping, aching, throbbing, and heavy pain (**Figure 1** [↗](#), plot B). Raw time courses of pain ratings during CPA are displayed in **Figure 1** [↗](#) (plot D), from which mean pain during the 4 minutes of stimulation was calculated (**Table 2** [↗](#), **Figure 1** [↗](#), plot F). No participants were excluded from the analyses of pressure pain (nicotine: $n = 29$; placebo: $n = 33$).

Estimates from the difference score 2W-LCS model for CPA pain ratings (**Figure 4** [↗](#)) for total ($b = -0.28$, $p = .50$, bootstrapped 95% CI: [-1.13, 0.49]), direct ($b = -0.17$, $p = .69$, bootstrapped 95% CI: [-1.07, 0.59]), and indirect ($b = -0.11$, $p = .53$, bootstrapped 95% CI: [-0.61, 0.18]) effects of nicotine on CPA ratings were non-significant (model fit: $\chi^2(24) = 29.55$, $p = 0.20$, CFI = 0.97, RMSEA = 0.061, SRMR = 0.085). Therefore, there was no evidence to suggest that nicotine gum directly or indirectly influenced change in CPA ratings.

Consistent with the PHP model, there was a significant effect of nicotine gum on change in global PAF (8–12 Hz) speed ($b = 0.089$, $p = .011$, bootstrapped 95% CI: [0.018, 0.16]), where change in PAF speed increased by 0.089 for the nicotine group compared to the placebo group. Additionally, there was again an effect of sex on change in PAF ($b = -0.076$, $p = .027$, bootstrapped 95% CI: [-0.15, -0.01]), where the increase in PAF speed was greater for females than males. And the estimated effect of change in PAF speed on change in pain ratings was non-significant for CPA ($b = -1.28$, $p = .49$, bootstrapped 95% CI: [-5.21, 2.11]).

Exploratory analysis of interactions between time point (i.e. pre-/post-gum) and group (i.e. nicotine/placebo) on CPA ratings using RM-ANOVA (Supplementary Material 5) suggested that CPA pain ratings decreased after chewing gum irrespective of group (**Figure 1** [↗](#), plots D and F).

Exploratory cluster-based permutation analysis

The effect of nicotine on PAF speed was only significant for global PAF with a wide frequency window (8–12 Hz) and not for sensorimotor PAF (i.e. Cz, C1, C2, C3, C4) or narrow frequency windows (9–11 Hz; Supplementary Materials 3.4–3.6). This suggests that nicotine produces change in PAF somewhere in sensor-space, but not in the pre-defined sensorimotor region comprising central electrodes. An exploratory cluster-based permutation analysis (see Methods) showed significant increases in PAF speed after chewing nicotine gum, with this effect being maximal over two spatially separated clusters of electrodes: one cluster over the central-parietal region, and another over the right-frontal region (**Figure 2** [↗](#), plot F). None of the electrodes exceeded the threshold of .05 to be clustered in the placebo group.

The potential mediating effect of this change in PAF on change in PHP and CPA was explored (not pre-registered) by averaging within each cluster (central-parietal: CP1, CP2, Cpz, P1, P2, P3, P4, Pz, POz; right-frontal: F8, FT8, FT10) and across both clusters. This averaging across electrodes produced three new variables, each assessed in relation to mediating effects on PHP and CPA ratings. The resulting in six exploratory mediation analysis (difference score 2W-LCS) models

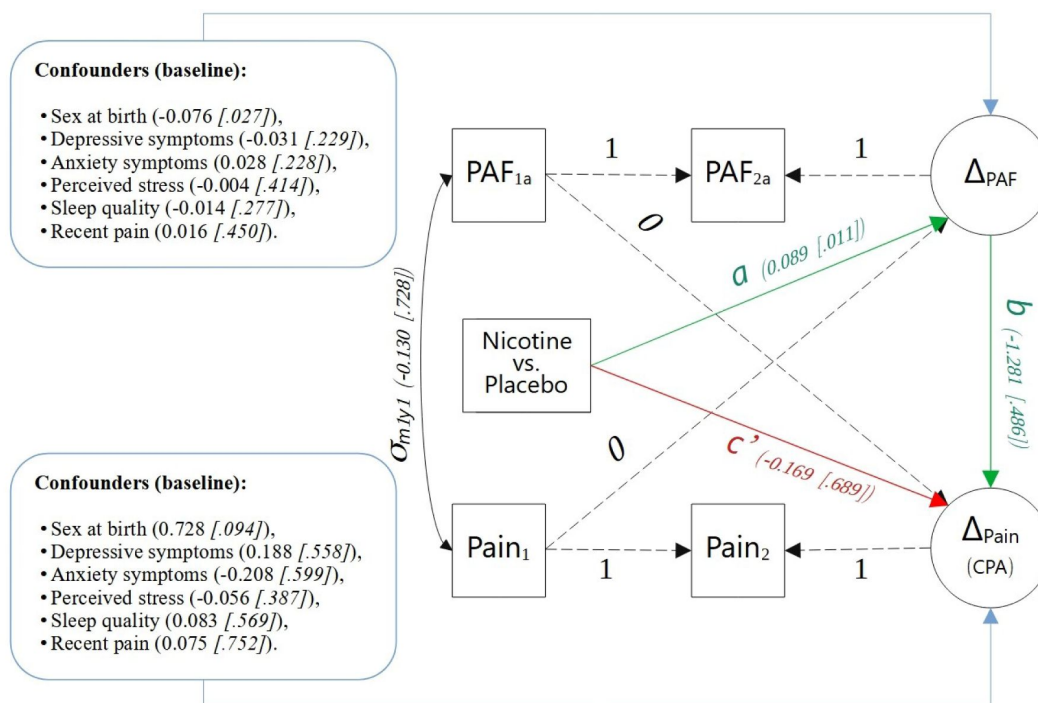


Figure 4

Path model displaying unstandardised estimates and p-values for the difference score two-wave latent change score (2W-LCS) model assessing the effect of nicotine against placebo (i.e. gum group) on change in global 8–12 Hz peak alpha frequency (i.e. Δ_{PAF}) and change in mean cuff pressure algometry (CPA) ratings (i.e. Δ_{Pain}). Green arrows highlight the indirect effects (i.e. a and b), the red arrow highlights the direct effect (i.e. c'), the blue lines highlight pre-determined confounding variables, and dashed lines denote fixed effects. This model suggests that chewing nicotine gum, compared to placebo, increases the speed of PAF and does not influence CPA ratings, directly or indirectly.

demonstrated minimal differences from the main analysis of global PAF (8–12 Hz), except for the expected stronger effect of nicotine on change in PAF ($bs = 0.11$ – 0.14 , $ps < .003$; Supplementary Materials 3.8–3.10).

Summary of mediation analyses

In summary, nicotine demonstrated significant total and direct effects on pain intensity ratings (i.e., reduction compared to placebo) during PHP, but not during CPA, where pain ratings decreased irrespective of gum group. There was also a significant effect of nicotine on change in global PAF speed (i.e., increased compared to placebo). However, there was no evidence of an indirect effect, suggesting that the change in PAF speed did not mediate the change in pain intensity ratings during PHP.

2.3 Relationship between PAF and pain at baseline

Evidence for negative correlations between pre-gum PAF (8–12 Hz) and pre-gum pain ratings during PHP ($r = -0.14$, $BF_{10} = 0.52$) and during CPA ($r = -0.086$, $BF_{10} = 0.33$; **Figure 5**) is inconclusive. Therefore, there was no evidence for or against slower PAF being associated with higher pain ratings. However, in an exploratory analysis (not pre-registered) of separate correlations in males and females (**Figure 5**, plot C) akin to those conducted in previous research on this topic [10], there was moderate evidence for a negative relationship between global PAF (8–12 Hz) and PHP ratings in males ($r = -0.39$, $BF_{10} = 3.22$). In contrast, the evidence for a relationship between global PAF and PHP ratings in females was equivocal ($r = -0.032$, $BF_{10} = 0.40$). Separating by sex for the correlations between PAF and CPA ratings still produced inconclusive evidence for a relationship (males: $r = -0.23$, $BF_{10} = 0.61$; females: $r = -0.086$, $BF_{10} = 0.40$).

The relationship between PAF and pain was also assessed using median splits on PAF, showing similar patterns of slower PAF being associated with higher PHP ratings in males only (Supplementary Material 6.1). Correlations between global PAF (8–12 Hz) and max pain during PHP also suggested slower PAF was only associated with higher max PHP ratings in male participants (Supplementary Material 6.2). Five participants could not tolerate the 46 °C stimulus during familiarisation, therefore a lower temperature of 45 °C was used for these participants. Repeat analyses excluding these five participants did not impact the observed relationships (Supplementary Material 6.3).

In summary, the evidence for a relationship between PAF and pain is inconclusive, with the exception of the relationship between PAF and PHP ratings by sex, where there was moderate evidence of a negative correlation in male participants.

3 Discussion

In this pre-registered study, we probed the proposed PAF–pain relationship [8–12] by administration of nicotine – a substance suggested to acutely alter both PAF [46] and pain [51]. Compared to placebo, our data demonstrate three key findings: 1) nicotine increased PAF speed; 2) nicotine reduced prolonged heat pain intensity but not prolonged pressure pain intensity compared to placebo gum; and 3) decreases in prolonged heat pain intensity were not mediated by changes in PAF. This suggests that changes in heat pain due to nicotine occur through mechanisms unrelated to change in PAF. In addition, we found moderate evidence that 4) slower PAF speed at baseline is associated with higher heat pain ratings at baseline in males but not females. In contrast PAF was not associated with pressure pain ratings for males or females.

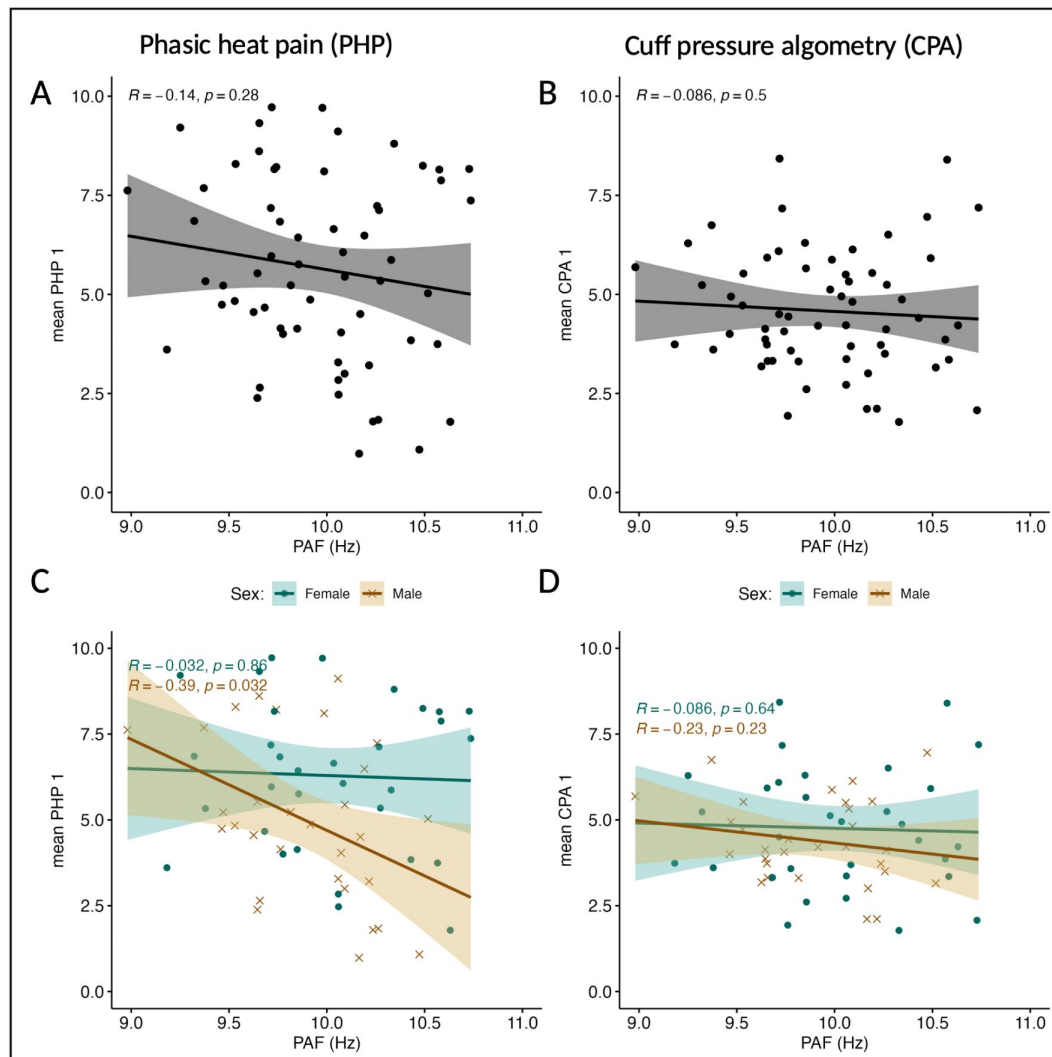


Figure 5

Wide band (8–12 Hz) global peak alpha frequency (PAF) does not correlate with A) mean pain ratings during phasic heat pain (PHP) or B) cuff pressure algometry (CPA). Regression lines and shaded 95% confidence intervals. Correlations separated by sex for females in teal ($n = 32$) and males in brown ($n = 30$) for C) PHP and D) CPA, suggest a strengthening of a negative relationship between sensorimotor PAF and PHP ratings for males ($BF_{10} = 3.22$).

3.1 The effect of nicotine on PAF

We aimed to determine whether nicotine gum could increase PAF speed. In line with previous literature in non-smokers [38,47,48], we contribute additional evidence that chewing 4 mg of nicotine gum increases PAF speed in non-smoking nicotine gum naïve, pain-free, adult participants. A cluster-based permutation analysis indicated that the change in wide window PAF (8–12 Hz) produced by nicotine, was most pronounced in a cluster of electrodes in the central-parietal region and a separate cluster of electrodes in the right-frontal region, rather than in the pre-registered sensorimotor ROI (i.e., Cz, C1, C2, C3, C4).

Interestingly, in addition to effects of nicotine on PAF, the 2W-LCS mediation model suggested that sex also significantly influenced change in PAF, with greater increases in PAF for females compared to males. Notably, the female participants in this study were significantly lighter than the males (Supplementary Material 3.4). However, since the 2W-LCS model did not examine the potential interaction between sex and gum group on change in PAF, conclusions are limited at this stage and future research should investigate these interactions.

3.2 The effect of nicotine on pain

Short-term nicotine use is thought to have acute analgesic properties in experimental settings, with a review reporting that nicotine increased pain thresholds and pain tolerance [51]. In addition, research in a rat model suggests analgesic effects on mechanical thresholds after short-term nicotine use [69]. However, previous research has not assessed the effects of nicotine on prolonged experimental pain models. The present study found that 4 mg of nicotine reduced heat pain ratings during prolonged heat pain compared to placebo for our human participants, but that prolonged pressure pain decreased irrespective of which gum was chewed. Our findings are thus partly consistent with the idea that nicotine may have acute analgesic properties [51], although further research is required to explore factors that may influence nicotine's potential impact on a variety of prolonged pain models. We further advance the literature by reporting this effect in a model of prolonged heat pain, which better approximates the experience of clinical pain than short lasting models used to assess thresholds and tolerance [51].

However, we note that the observed effect of nicotine on pain was small in magnitude, and most prominent in comparison to the effect of placebo, where pain ratings increased after chewing, which brings into question whether this reduction in pain is meaningful in practice. While acknowledging the modest effect size, it's essential to consider the broader context of our study's focus. Assessing the clinical relevance of pain reduction is pertinent in applications involving the use of any intervention for pain management [70]. However, from a mechanistic standpoint, particularly in understanding the implications of and relation to PAF, the specific magnitude of the pain effect becomes less pivotal. Nevertheless, future research should examine whether effects on pain increase in magnitude with different nicotine administration regimens (i.e. dose and frequency).

We did not see an analgesic effect for prolonged pressure pain, as ratings during pressure pain were lower at the second assessment irrespective of gum allocation. The way we calibrated the pressure pain model may have impacted our results. Previous literature has used cuff inflation where the cuff pressure increases until the participant reports 4/10 pain, at which point it remains stable for the desired number of minutes [59–61,71]. In contrast, a separate calibration was conducted at the beginning of this experiment to calculate the pressure used for the remainder of the study, based on the average of three trials of increasing pressure to 4/10 pain (see Methods). This method may result in lower pain ratings at the second pain induction with the same pressure in a same-session repeated measures investigation.

Analgesic mechanisms of nicotine are complex [51], likely depending on the type of pain and whether nicotine use is acute or chronic, along with several other contributing factors. For example, in addition to the effect of nicotine on prolonged heat pain ratings, our results suggest an effect of stress on changes in heat pain ratings, with those self-reporting higher stress at baseline having greater reductions in pain. Our post-hoc analysis suggested that this relationship between higher stress and larger decrease in PHP ratings was only present for the nicotine group (Supplementary Material 3.3). As stress is linked to nicotine use [72,73] and pain [74–76], these interactions should be explored in future.

3.3 The relationship between PAF and pain

Prior research suggests that slower PAF is associated with higher pain sensitivity [8–11], and while the validation of PAF as a predictive biomarker continues [14], the question of whether PAF is mechanistically linked to pain remains unanswered. Modulating PAF speed and pain sensitivity with nicotine is necessary but not sufficient to demonstrate a causal pathway from nicotine to PAF to pain (i.e. intervention → mediator → outcome) [77]. We did not find evidence for a causal relationship between PAF and pain sensitivity, as our 2W-LCS model did not show a significant mediating (i.e. indirect) effect of change in PAF on change in pain due to nicotine. The implication of this is that pain sensitivity may not be dependent on state changes in PAF, thus changing PAF may not be a useful target for pain modulation. However, as this is the first study to directly manipulate PAF and observe effects on pain, we cannot confidently state that a causal relationship between PAF and pain sensitivity does not exist, and future research should explore these relationships with larger sample sizes, larger doses of nicotine, and alternative PAF modulators.

As an alternative explanation, an indirect effect may exist, but could not be seen in the present study due to the single, low dose of nicotine or for other methodological reasons. For example, the relationship between change in PAF and change in pain may be non-linear, and therefore difficult to see using a 2W-LCS model that is based on linear regression analyses. Additionally, the use of global PAF may have introduced mediation measurement error into our mediation analysis. The spatial precision used in the current study was based on previous literature on PAF as a biomarker of pain sensitivity, which have used global and/or sensorimotor ROIs [8,10]. Identification and use of the exploratory electrode clusters found in this study could build upon the current work (e.g., [78]). However, exploratory analysis of the clusters found in the present analysis demonstrated no influence on mediation analysis results (Supplementary Materials 3.8–3.10). Alternatively, independent component analysis (ICA) could be used to identify separate sources of alpha oscillations [79], as used in other experimental PAF–pain studies [8,18], which could aid to disentangle the potential relevance of different alpha sources in the PAF–pain relationship. Although this comes with the need to develop more reproducible and automated methods for identifying such components. In-house code was developed to compare a sensorimotor component to the results presented in this manuscript (Supplementary Material 3.11), showing similar results to the sensorimotor ROI mediation analysis presented here. However, examination of which alpha – be it sensor or source space – are related to pain, how they can be robustly represented, and how they can be manipulated are ripe avenues for future study.

Furthermore, there may have been other mediators that suppress the mediating effects of PAF within our model [66,80,81]. This is plausible, as nicotine is thought to act on pain through multiple pathways [52,54,82], and more direct PAF modulation could more definitively evaluate whether PAF is mechanistically linked to pain sensitivity. Future interventions to modulate PAF could use non-invasive brain stimulation (NIBS), such as transcranial alternative current stimulation (tACS) or repetitive transcranial magnetic stimulation (rTMS), to influence the firing of neurons, thereby directly influencing the brain mechanisms producing PAF [83–85].

Regarding the relationship between PAF and pain at baseline, the negative correlation between PAF and pain seen in previous work [7, 11, 16] was only observed here for male participants during the PHP model for global PAF in an exploratory analysis. Effect sizes for the relationship between PAF and pain ratings from previous work were in the range of $r = 0.4$ – 0.6 [8, 11]. In the present study we saw a weaker relationship for PHP pain ($N = 62$, $r = -0.14$). However, sex-stratified analysis demonstrated that males exhibit comparable effect sizes to previous studies ($n = 30$, $r = -0.39$), while females did not show a relationship between PAF and heat pain ($n = 32$, $r = -0.032$). Subtle variations in experimental design and stimulus delivery could have influenced the PAF–pain relationship. Crucially, the present study did not calibrate the temperature of the PHP model based on participants’ thresholds for moderate pain. Consequently, the fixed temperature did not produce a moderately painful stimulus for all participants, in contrast to previous studies that calibrated to a moderately painful stimuli [10]. This discrepancy may account for the correlation observed between PAF and heat pain ratings among only the male participants in our study, as fixed stimuli can create ceiling or floor effects [86]. Previous literature has also acknowledged the potential for sex differences in the PAF–pain relationship [17].

In parallel, we saw no indication of a relationship between PAF and pain ratings during CPA. The introduction of the CPA model, specifically calibrated to a moderate pain threshold, provides further support for the notion that the relationship between PAF and pain is specific to certain pain types [17, 28]. Prolonged heat pain was pre-dominantly described as moderate/severe shooting, sharp, and hot pain, whereas prolonged pressure pain was predominantly described as mild/moderate throbbing, cramping, and aching in the present study. It is possible that the PAF–pain relationship is specific to particular pain models and protocols [12, 17].

Previous experimental pain literature has suggested that slower PAF is related to higher pain sensitivity for PHP [10], capsaicin heat pain [8, 10], and nerve growth factor (NGF) [9, 13] models, while faster PAF has been related to higher pain sensitivity in tonic heat pain [7] and NGF [15]. Here we contribute to this literature by showing no relationship between baseline PAF and pain sensitivity during the CPA model. Further investigation of the PAF–pain relationship with a variety of prolonged pain models is warranted [12].

3.4 Study limitations and constraints

This study presents opportunities for future research due to certain limitations. First, the dose of nicotine administered was equivalent to one cigarette [87]; however, to enhance the efficacy of this research, exploring higher doses or dosing based on weight could be beneficial. Second, blood or saliva samples were not collected in the present study, therefore it is unclear whether participants absorbed similar amounts of nicotine. Future studies in this area should also be powered to perform sex-disaggregated analyses, as our findings indicate a relationship between baseline PAF and subsequent heat pain intensity only in males, and stronger nicotine modulation of PAF in female participants. The ability to detect changes in PAF can be considerably impacted by the frequency resolution used during Fourier Transformations, an element that is overlooked in recent methodological studies on PAF calculation [16, 88]. Changes in PAF within individuals might be obscured or conflated by lower frequency resolutions, which should be considered further in future research.

3.5 Conclusion

In this study, we employed a rigorous double-blind, randomised, placebo-controlled design, pre-registered the analysis, and introduced a novel difference score 2W-LCS mediation model to examine the mediating role of changes in PAF on pain while controlling for pre-determined confounders. These methodological choices allowed for a comprehensive investigation into whether PAF plays a mechanistic role in pain sensitivity.

We have shown that, compared to placebo, nicotine gum increases global PAF speed and decreases pain ratings during prolonged heat pain. However, the effects of nicotine on heat pain were not mediated by changes in PAF. Therefore, we find no evidence that PAF is causally related to pain. The implication of this is that changing PAF may not be a useful target for pain modulation, because pain sensitivity may not be dependent on state changes in PAF. However, this is the first study to test modulation of PAF as a mechanism for acute analgesia. More randomised experiments of the effects of nicotine that use larger sample sizes, track the pharmacokinetics of nicotine, whilst reducing mediator measurement error are needed to uncover whether these factors account for the lack of mediation effect seen here.

A growing body of evidence indicates a relationship between PAF and pain, including chronic pain studies [17,24,26–28,89], and experimental pain research from our own [8–12,14] and other groups [7,15–18]. Alongside validation of its clinical predictive utility [14,88], further research is needed to ascertain the causal nature of the relationship, explore PAF as a potential target for pain modulation, and gain deeper mechanistic insights into the PAF–pain relationship.

4 Methods

4.1 Design overview

The study employed a randomised, double-blind, placebo-controlled, parallel experimental design. Testing sessions were balanced for treatment (i.e. nicotine or placebo gum) and stratified by sex. Individuals were required to attend one 150 minute session. The experimental flow for this study is summarised in [Figure 6](#).

4.2 Participants

From June 2021 to March 2022, 62 individuals participated at Neuroscience Research Australia (NeuRA). Inclusion criteria for the study were for participants to be healthy, pain free, non-smoking, and between 18–45 years of age, inclusive.

Non-smokers were defined as individuals who have smoked fewer than 100 cigarettes and/or e-cigarettes in their lifetime and have not smoked over the past 12 months. Exclusion criteria: previous use of nicotine gum; history of neurological, musculoskeletal, cardiovascular, or psychiatric conditions; other major diseases or disorders; current muscle pain; breastfeeding or pregnant; use of opioids, illicit drugs, or excessive alcohol in the past three months; or unable to provide informed consent.

4.3 Ethics approval and consent to participate

The Human Research Ethics Committee (HREC) Executive at the University of New South Wales (UNSW) approved this project (HC190466).

4.4 Experimental protocol

Participants were screened for eligibility by telephone and were asked to refrain from drinking coffee or caffeinated drinks two hours prior to the start of testing, as caffeine may impact nicotine absorption [90,91].

After signing informed consent, cool and warmth detection tests were conducted, followed by PHP familiarisation, then CPA calibration. During these introductions the participants were also familiarised with the VAS pain rating.

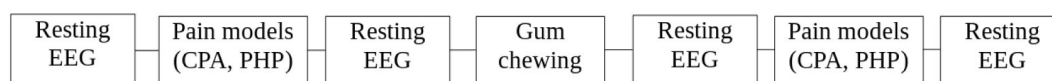


Figure 6

Timeline of experimental events. Electroencephalography (EEG) data were collected during eyes open and eyes closed resting states before and after each pain assessment. Prolonged pain assessments consisted of cuff pressure algometry (CPA) followed by phasic heat pain (PHP) models, and took place before and after gum chewing.

EEG capture started with the control condition of three cycles of warm temperatures, after which the remaining experimental flow is illustrated in **Figure 6**. The first resting state period was recorded, followed by the first pain assessments (i.e. CPA and PHP model), which were immediately followed by the second resting state. EEG recorded during the pain assessments was not analysed for the present article. Then either nicotine or placebo gum was administered with the chew procedure, followed by the third resting state, the second pain assessments, and the fourth resting state. The experiment ended with participants completing the gum chew experience questionnaire and debrief.

4.5 Data collection procedures

Questionnaires

Participants completed a set of questionnaires to assess demographics, medical history, general health, mental health, pain, and sleep. This data was collected electronically and stored using REDcap [92,93].

Questionnaires included the following National Institutes of Health common data elements (CDE) for pain biomarkers: Pain Catastrophizing Scale (PCS) [94]; Brief Pain Inventory (BPI), Pain Severity and 7-item Interference subscales [95,96]; SF-8 to assess general health [97]; Patient Health Questionnaire-2 (PHQ-2) to assess depression [98]; Generalised Anxiety Disorder Questionnaire-2 (GAD-2) to assess anxiety [99]; Pennebaker Inventory of Limbic Languidness (PILL) to assess reporting of physical symptoms and sensations [100]; Tobacco, Alcohol, Prescription medications, and other Substances (TAPS); sleep scale. The sleep scale asked “how would you rate your sleep last night?”, on a scale from 1 (i.e. “poor sleep quality”) to 10 (i.e. “excellent sleep quality”). In addition, a health history form, the Karolinska sleepiness scale (KSS) [101], Epworth sleepiness scale (ESS) [102], the perceived stress scale (PSS) [103], and questions regarding participants current level of weekly physical activity were included.

Electroencephalography

EEG was recorded from 63 active wet electrodes (actiCap, Brain Products GmbH, Germany), embedded in an elastic cap (EASYCAP, GmbH) in accordance with the international 10-10 system. Recordings were referenced online to ‘FCz’ and the ground electrode was placed on ‘FPz’. A sampling rate of 5000 Hz was used and impedance was kept below 25 kOhms throughout testing using electrolyte gel. EEG signal was amplified and digitised using an actiCHamp Plus DC amplifier linked to BrainVision Recorder software (version 1.22.0101, Brain Products GmbH, Gilching, Germany).

For each recording, resting state EEG was obtained during eyes open (three minutes) followed by eyes closed (three minutes). The eyes open recording took place in low lighting while participants rested their gaze on a white fixation cross with a black background displayed on a monitor. The eyes closed recording took place with the lights off. For both recordings participants were asked to rest in a comfortable seated position and try not to fall asleep.

Pain assessment

Two tonic (i.e. prolonged) pain assessments were conducted during the study, a phasic heat pain (PHP) model [104] and a cuff pressure algometry (CPA) model [59,104]. Pain intensity was sampled at 1 Hz during pain stimulation using a visual analogue scale (VAS), with anchors ‘no pain’ on the left to ‘worst pain imaginable’ on the right, custom MATLAB (version R2021a) code (source available: <https://osf.io/pc4rq/>).

Participants were instructed to press the ‘STOP’ button on the screen if they became too uncomfortable and wanted to stop the pain assessments.

The Short Form McGill Pain Questionnaire 2 (SF-MPQ-2) [105] was used after each pain model to assess experienced qualities (e.g., throbbing, cramping, burning).

Heat pain

Equipment. Thermal stimuli were applied to the volar surface of the left forearm using a thermal-contact heat stimulator (27 mm diameter Medoc Pathways CHEPS Peltier device, Medoc Advanced Medical Systems Ltd., Ramat Yishai, Israel).

Cool and warmth detection thresholds. The thermode was placed on the volar surface of the left arm at a neutral temperature (32 °C). Participants were instructed to press a button held in their right hand when they sensed a change in temperature. Four areas were marked out by drawing three perpendicular lines equidistant from the elbow crease to 4 cm below the base of the palm. On each trial, the researcher moved through these locations on the forearm in a randomly generated sequence. There were three trials each for warmth (ascending temperatures) and cool (descending temperatures) detection thresholds. Participants were required to consistently detect the warmth and cool transitions in order to continue with the experiment.

Heat pain threshold. Participants were required to press the hand-held button with their right hand the moment they felt any pain as the thermode temperature increased at a rate of 1.5 °C/second. This pain threshold test was conducted three times and an average heat pain threshold was calculated.

Familiarisation. Participants were initially introduced to the PHP model (see below) by receiving two cycles of the heat pain, to ensure they would tolerate the maximum temperature of 46 °C during the experiment and to familiarise participants with using the VAS. If a participant was uncomfortable with receiving 46 °C, the temperature was decreased by 1 °C and the PHP familiarisation was repeated to ensure this temperature was tolerable.

Phasic warmth. Participants experienced three rounds of a phasic warm stimulus. Stimulus trains consisted of a rise phase of 2 °C/s, followed by a 40 second phase where a temperature was held at 36 °C, then a fall phase at a rate of 1 °C/s to reach 32 °C. This neutral temperature was then held for 20 seconds before the cycle repeated three times.

Phasic heat pain (PHP) model. The PHP model consisted of a series of five repetitions of phasic stimuli, each lasting one minute 21 seconds. Stimulus trains consisted of a rise phase of 2 °C/s, followed by a 40 second phase where a temperature was held at 46°C, then a fall phase at a rate of 1 °C/s to reach 32 °C. This neutral temperature was then held for 20 seconds before the cycle repeated five times. The PHP model lasted six minutes 45 seconds, and participants rated their pain continuously on the VAS. Similar methodology using repetitive-phasic models have shown that this procedure reliably evokes tonic pain and sensitisation in healthy participants [8,10].

Pressure pain

Equipment. Pressure was applied using a standard double-chamber adult arm cuff (length 45cm, width 15cm) placed in the middle of the left lower leg over the head of the gastrocnemius soleus muscle.

Familiarisation and calibration. Participants underwent a pressure cuff calibration. Similar to previous literature [60,62], the pressure was gradually increased manually (30 mmHg/5 seconds) until the participant reported a moderate pain intensity of 4/10 on the VAS and pressure was automatically recorded. This gradual pressure increase was conducted three times with a two

minute break between each test, to establish the average pressure required to evoke 4/10 pain for each participant. The maximum pressure limit for this study was 100 kPa (750 mmHg) [63]. This is well within safe limits, as previous studies have used maximums of 180 or 200 kPa without adverse effects [59–61].

Cuff pressure algometry (CPA) model. Pressure was increased manually (30 mmHg/5 seconds) until the individualised pressure was reached, then held constant for four minutes. Participants rated their pain continuously using the VAS.

Gum chewing

Nicotine was administered orally in the form of 4 mg of cinnamon flavoured Nicorette® gum (Johnson & Johnson Inc., Markham, Ontario, Canada); comparable to smoking a single cigarette, this produces a blood concentration of approximately 15–30 mg/ml [87]. The placebo gum was cinnamon flavoured Xyloburst® natural chewing gum, which, other than containing no nicotine, is of a similar flavour, size, and texture to the nicotine gum. At the end of the experiment, before being debriefed, participants completed a short questionnaire about the gum chewing experience to test perceptual matching between the nicotine and placebo gums.

To promote nicotine absorption, participants rinsed their mouth with a glass of water for 10 seconds prior to receiving the gum, as performed previously [90]. We used a protocol similar to Bowers and colleagues [38], following manufacturer's guidelines of a total chewing time of 25 minutes. Audio cues to bite the gum occurred every 30 seconds for a total of 25 minutes; participants were asked to alternate between the left and right sides of the mouth, and to park the gum between the teeth and cheek between chews. Participants watched a wildlife documentary during the chewing to maintain attention; all participants watched the same documentary to control for its possible effects.

Randomisation and blinding

Participants were randomised in a 1:1 ratio to receive either nicotine gum or placebo gum. The allocation list was created a priori, independently of the primary experimenter, using blocked randomisation based on sex at birth. The allocation was blind to both the participant and experimenter. All participants were told that they were receiving nicotine gum to provide equal expectations. Participants were debriefed at the end of the experiment and asked whether they thought they had been given nicotine, placebo, or if they weren't sure. Participants were unblinded by the independent researcher separately. The primary experimenter remained blinded with dummy variables until completion of analysis.

4.6 Data processing

EEG pre-processing

Blinded preprocessing of de-identified EEG was conducted with custom code in MATLAB 2020a (v.9.8.0.1417392, The MathWorks, Inc., Natick, MA, USA) using EEGLAB 2020.0 [106] and FieldTrip (v.20200607) [107].

Using EEGLAB, for four resting states, each participant's data were downsampled to 500 Hz, filtered between 2 and 100 Hz using a linear finite impulse response (FIR) bandpass filter, and the three minute periods of eyes closed resting state were visually inspected and overtly noisy channels with ≥ 8 signal deviations (e.g., popping, baseline drift, line noise) were removed. The average number of channels removed were 0.94, 0.98, 0.74, and 0.87 (range: 0–4) for the four resting states, respectively. Data from the remaining channels were re-referenced to the common

average reference. Removed channels were not interpolated for the pre-registered global and sensorimotor ROI averaged analyses, but were interpolated for an exploratory cluster based permutation analysis using the nearest neighbour average method in Fieldtrip.

Using FieldTrip, data were segmented into 5 second epochs, with no overlap, visually inspected using *ft_databrowser()*, and epochs containing any electrode, muscle, or motion artefacts were removed. From a maximum of 36 epochs (i.e. 3 mins), the average number of remaining epochs was 31.21, 31.33, 32.13, and 31.39 (range: 24–36) for each resting state, respectively. No resting state contained less than two minutes of data [88]. Principal component analysis using the *runica* method was applied to identify and remove components reflecting eye blinks and/or saccades. The average number of components removed per resting state was 1.05, 1.2, 1.1, 1.2 (range: 0–2), respectively.

Quantification of peak alpha frequency (PAF)

Each participant's PAF was estimated for the four eyes closed resting states. FieldTrip was used to calculate the power spectral density in the range of 2–40 Hz for each epoch in 0.2 Hz bins. A Hanning taper was applied to the data prior to calculating spectra to reduce any edge artefacts [108]. The PAF for each power spectral density was estimated using the centre of gravity (CoG) method [109–113]:

$$\text{CoG} = \frac{\sum_{i=1}^n f_i a_i}{\sum_{i=1}^n a_i} \quad (1)$$

where f_i is the i^{th} frequency bin, n is the number of frequency bins within the frequency window (8–12 Hz or 9–11 Hz), and a_i is the amplitude of the frequency for f_i . This equation states that PAF is the weighted sum of spectral estimates [109]. Global (i.e. average of all electrodes) and a sensorimotor region of interest were used (i.e. average of C3, C1, Cz, C2, and C4). PAF estimates for each participant at each time point were calculated. Out of 248 resting states recorded, 14 resting states had 4 ROI channels instead of 5. Importantly, no resting state had fewer than 4 channels for the sensorimotor ROI.

Contrary to some previous studies that used ICA to isolate sensory region alpha sources [8, 15, 18], we used pre-determined sensor level ROIs to improve reproducibility and reduce the potential for bias when individually selecting ICA components. Using sensor level ROIs may decrease the signal-to-noise ratio of the data; however, this approach has still been effective for observing the relationship between PAF and experimental pain [9, 10].

The frequency range used to calculate PAF varies in the PAF–pain literature. The narrower 9–11 Hz band reduces the impact of 1/f noise (i.e. trend of higher power at lower frequencies) on the PAF estimations [8, 108], however the narrow band may not produce representative PAF values for those with peaks near the lower or upper limits. Therefore, the wider 8–12 Hz band was used as the primary band of interest (Figure 2, plot A). All analyses were repeated with the narrower band and are displayed in Supplementary Material 3.4 and 3.6. Power was calculated using the wider 8–12 Hz band to display alpha on topographical plots.

4.7 Statistical analysis

Statistical analysis was conducted in RStudio (RStudio Team, 2021.9.1.372). A Guideline for Reporting Mediation Analysis (AGReMA) was followed for this article and the analysis plan was pre-registered (<https://osf.io/pc4rq/>).

Mediation analysis

Using a structural equation modelling approach, a two-wave latent change score (2W-LCS) model – recommended by Valente and MacKinnon [64] – was used to assess whether nicotine changed PAF or pain (i.e. hypotheses 1–3).

Prior to modelling, a paired equivalence test satisfied that PAF_{1a} (i.e. pre-gum before the first pain assessment) and PAF_{1b} (i.e. pre-gum after the first pain assessment) were equivalent ($t(60) = 0.90$, $p < .001$), therefore PAF_{1a} was used in the below models.

Latent change score (LCS) specification. Calculating the absolute difference between pre- and post-gum variables can result in regression to the mean [64]. With LCS specification, the second wave of PAF and pain (i.e. post-gum) can be set as a unit-weighted function of pre-gum and a new unobserved change variable (i.e. Δ_{PAF} and Δ_{pain}) to simulate subtracting pre-gum scores from post-gum scores on PAF and pain variables [64].

$$\Delta_{PAF} = PAF_{2a} - PAF_{1a} \quad (2)$$

$$\Delta_{pain} = Pain_2 - Pain_1 \quad (3)$$

Simulation of the above equations was achieved via the following parameter constraints. Loading on $Pain_2$ was fixed to one, while the mean and variance of Δ_{pain} were freely estimated. The path from $Pain_1$ to $Pain_2$ was fixed to 1, the mean (i.e. intercept) and variance of $Pain_2$ were constrained to zero, and the mean and variance of $Pain_1$ were freely estimated. The same approach was taken for Δ_{PAF} . The covariances between PAF_{1a} and $Pain_1$ were freely estimated.

Two-wave latent change score (2W-LCS) models. The above latent change variables (Equation 2 and Equation 3) can then be used within 2W-LCS models to conduct mediation analysis. Two model variations were constructed to assess total, direct, and indirect effects of nicotine separately for latent change in mean ratings during PHP and CPA as outcomes (Table 4). Models were constructed in R (version 4.2.1) using the lavaan package [116] with maximum likelihood estimation and 1000 bootstraps (seed: 2022) to compute standard errors. The mediation model syntax is available online (<https://osf.io/pc4rq/>).

The ANCOVA-equivalent 2W-LCS model. The first model variation used was the ANCOVA-equivalent model [64], the path diagram is displayed in Figure 7 and the below equations were used (Equation 4 and Equation 5):

$$\Delta_{PAF} \sim i_1 + a_{m2x}Gum + s_{m2m1}PAF_1 + b_{m2y1}Pain_1 + \text{covariates} + e_1 \quad (4)$$

$$\Delta_{pain} \sim i_2 + c'_{y2y1}Gum + s_{y2y1}Pain_1 + b_{y2m1}PAF_1 + b_{y2m2}\Delta_{PAF} + \text{covariates} + e_2 \quad (5)$$

In this model, i denotes the intercept and e denotes error. Both $s_{m2m1}M_1$ and $s_{y2y1}Y_1$ are time-lagged autoregressions referring to change in M and Y due to prior levels of M and Y , respectively. These autoregressions reflect that each variable influences itself over time. The $b_{y2m1}M_1$ and $b_{m2y1}Y_1$ are cross-lagged coupling parameters, which refer to effects of previous levels of M on change in Y and effects of previous levels of Y on change in M . In other words, each variable influences the other variable at the subsequent time point. As displayed in Figure 7, X is predicting constant change in Y via the direct c' -path (Equation 5; c') and constant change in M via the a -path (Equation 4; a_{m2x}). Latent change in M is predicting later latent change in Y via

Model	Intervention	Mediator	Outcome
1.1	Nicotine	Latent Δ PAF	Latent Δ mean heat pain
1.2	Nicotine	Latent Δ PAF	Latent Δ mean pressure pain

Table 4

Overview of mediation models.

the constrained b -path (Equation 5; b_{y2m2}). The indirect effect was assessed by estimating the product of coefficients ab ($a_{m2x'} * b_{y2m2}$). The total effect was assessed by summation of the indirect and direct effects (i.e. $c + [a_{m2x'} * b_{y2m2}]$).

Difference score 2W-LCS model. We compared the ANCOVA-equivalent model (Figure 7) to the difference score model (Figure 8); the former adjusts for the effects of pre-gum PAF and pain on post-gum PAF and pain, while the latter assumes those paths are equal to zero [64]. Removal of autoregressive (i.e. $s_{m2m1}M_1$ and $s_{y2y1}Y_1$) and coupling parameters (i.e. $b_{y2m1}M_1$ and $b_{m2y1}Y_1$) produces a simpler unconditional model of change [65,117]. The path diagram is displayed in Figure 8 and the below equations were used (Equation 6 and Equation 7):

$$\Delta_m \sim i_3 + aGum + \text{covariates} + e_3 \quad (6)$$

$$\Delta_y \sim i_4 + c'Gum + b\Delta_{PAF} + \text{covariates} + e_4 \quad (7)$$

Model fit and comparison. Model fits were compared with χ^2 statistic ($p > 0.05$), comparative fit index (CFI), standardised root mean squared residual (SRMR), and root mean square error of approximation (RMSEA) [65,118]. For adequate model fit: CFI > 0.90, RMSEA < 0.08, and SRMR < 0.08 [119].

Confounding. The intervention-mediator (a -path) and intervention-outcome (c' -path) relationships are unconfounded due to the random allocation of nicotine or placebo gum. For the mediator-outcome (b -path) relationship, potential confounders were chosen by selecting variables measured prior to gum allocation that we assume to be a cause of the mediator, outcome, or both [120].

We assume that the mediator-outcome relationship is unconfounded after adjusting for the pre-determined (Supplementary Material 7.1; [121,122]) minimum sufficient set of observed confounders: sex at birth, depression symptoms, anxiety symptoms, perceived stress, sleep quality, and recent pain (Figure 7). Within the model, the variance of each covariate was freely estimated, along with their co-variance with baseline values of PAF and pain.

Assumptions. Multicollinearity assessment (R^2 cut-off of 0.7) did not suggest any covariates requiring removal. Using the *mvn()* function, we found violation of multivariate normality (Mardia skewness = 361.28, $p < .001$; Mardia kurtosis = 2.39, $p = 0.017$) as well as univariate normality for covariates 2, 3, 5, and 6 ($W = 0.60$ - 0.93 , $p < 0.0019$), the impact of which was reduced by use of bootstrapping [123,124]. There was no systematically missing data and only complete datasets were used, incomplete sets were removed by listwise deletion.

Mediation using PROCESS. PROCESS is a statistics package in R for conducting mediation analysis [66], and is frequently used in the literature [125–127]. PROCESS analysis was conducted using an ANCOVA-equivalent model without the LCS specification, as construction of latent variables is not currently possible in this package [66]. Pain and PAF post-gum (i.e. PAF_{2a} and $pain_2$) were used as outcome measures rather than change in each, with PAF_1 and $pain_1$ included alongside the other covariates. The estimates produced by PROCESS were similar to the ANCOVA-equivalent 2W-LCS model in lavaan, and are displayed in the Supplementary Material 7.2.

Cluster-based permutation analysis. An exploratory non-parametric cluster-based permutation analysis [128,129] was conducted to assess where changes in global PAF occurred in sensor space, similar to those conducted by Furman and colleagues [78]. Tests conducted at the single electrode level were thresholded at $p < .01$. If two or more spatially adjacent electrodes passed this threshold, they were grouped into a cluster, and each electrode within a cluster had to be adjacent to at least one other electrode from the overall cluster. Following the formation of clusters, test statistic values were summed across all electrodes within a cluster to obtain a composite test score

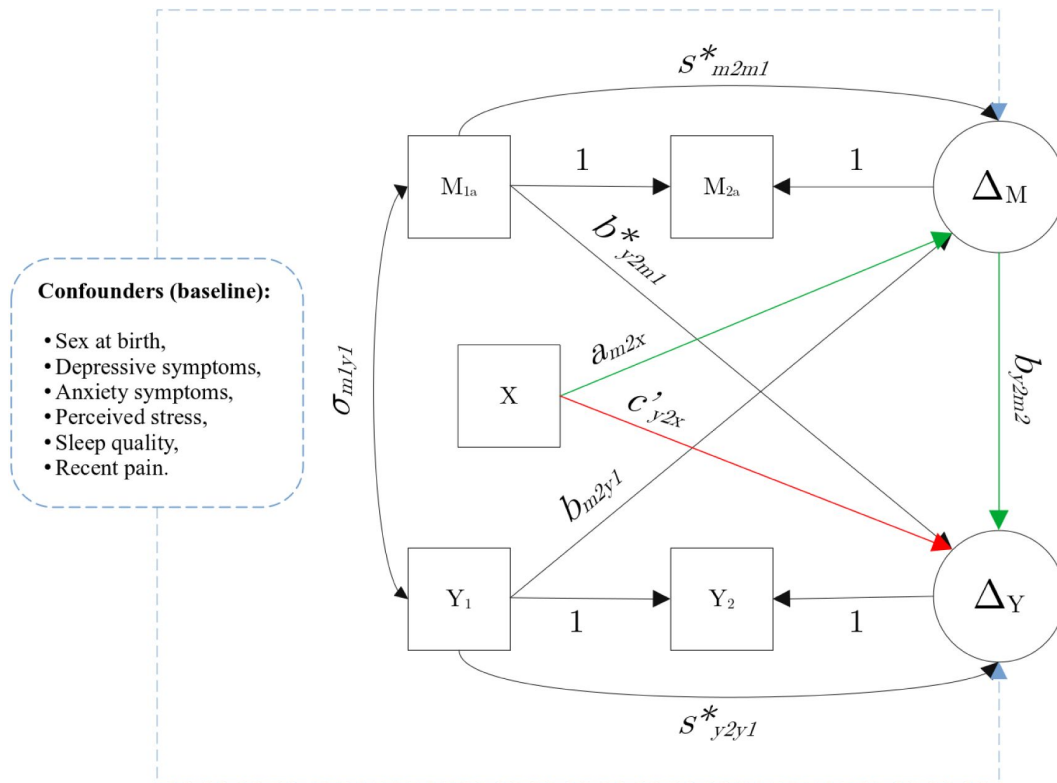


Figure 7

Adapted from Valente and MacKinnon [64]. The analysis of covariance (ANCOVA) equivalent pre-test/post-test control group design (two-wave) with latent change score (LCS) specification for the outcome (Y: pain) and mediating (M: PAF) variables. The direct effect of the intervention on the outcome is represented by the red line (c'). The intervention-mediator relationship is represented by a green line (a_{m2x}) from the intervention (i.e. X: nicotine vs. placebo) to the mediator (i.e. Δ_M). The mediator-outcome relationship is represented by another green line (b_{y2m2}) from the mediator (i.e. Δ_M) to the outcome (i.e. Δ_Y). Together the two green lines represent the indirect effect of the intervention on the outcome via the mediator. Time-lagged autoregression paths are represented by S^*_{m2m1} and S^*_{y2y1} . Cross-lagged coupling paths are represented by b_{m2y1} and b^*_{y2m1} . The covariance between Y and M at baseline is represented by σ_{m1y1} . The potential confounders of the mediator-outcome relationship were measured at baseline and are represented by the blue dashed lines. *Note.* * = latent; X = intervention (i.e. gum); M = mediator (i.e. PAF); Y = outcome (i.e. pain).

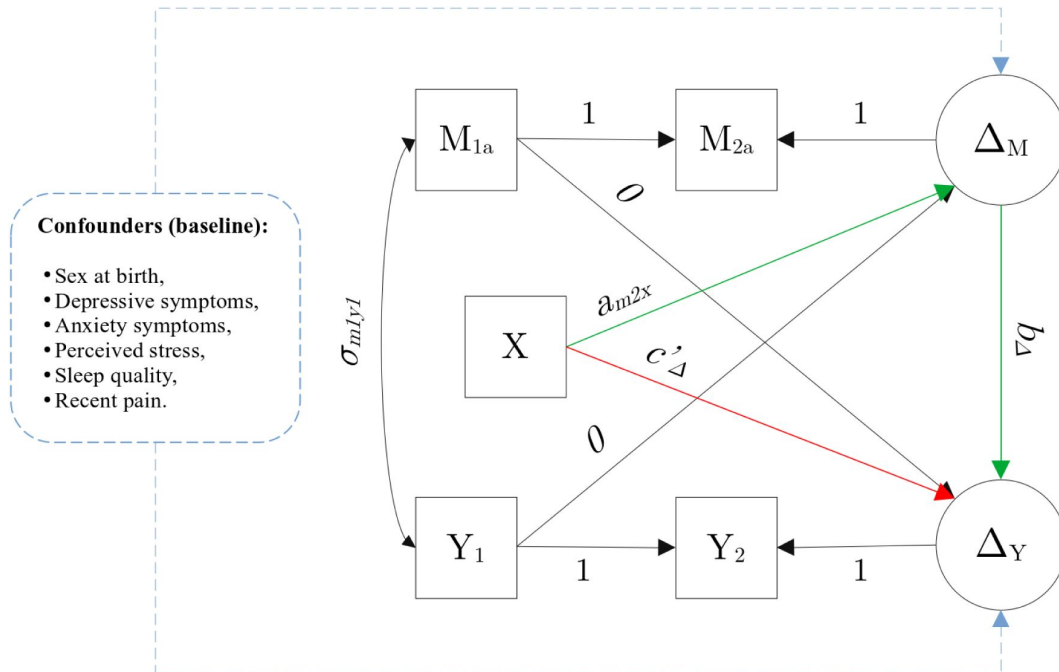


Figure 8

Adapted from Valante and MacKinnon [64]. The difference score model with latent change score specification for the outcome (Y) and mediating (M) variables. The direct effect of the intervention on the outcome is represented by the red line (c'). The intervention-mediator relationship is represented by a green line (a_{m2x}) from the intervention (i.e. X: nicotine vs. placebo) to the mediator (i.e. Δ_M). The mediator-outcome relationship is represented by another green line (b_Δ) from the mediator (i.e. Δ_M) to the outcome (i.e. Δ_Y). Together the two green lines represent the indirect effect of the intervention on the outcome via the mediator. Cross-lagged coupling paths are constrained to 0 and time-lagged autoregressions are constrained to 1. The covariance between Y and M at baseline is represented by σ_{m1y1} . The potential confounders of the mediator-outcome relationship are represented by the blue dashed lines. *Note.* X = intervention (i.e. gum); M = mediator (i.e. PAF); Y = outcome (i.e. pain).

for that cluster. For each potential cluster, this composite test score was then compared against a null distribution of 1000 generated scores produced by random shuffle of values for the dependent variable across conditions and participants at each electrode of the cluster. If a cluster's composite test statistic surpassed the 99th percentile of the null distribution, that cluster was evaluated as significant.

PAF–pain relationship

To address whether faster sensorimotor or global PAF during pain-free resting states was associated with lower pain ratings during the first PHP and CPA models before gum chewing (i.e. hypothesis 4), correlations were used [7–10].

Bayesian correlations were conducted in R ('BayesFactor' package, default priors) between pain-free baseline 8–12 Hz PAF (i.e. global and sensorimotor) and pain measures (i.e. mean, max, AUC) during each model (i.e. PHP, CPA) prior to gum administration.

Median splits on PAF speed were conducted, and the difference in pain reported between PAF speed groups was assessed by independent t-tests for mean and AUC pain, with Mann Whitney-U test for max PHP pain, due to violation of normality ($W = 0.89$, $p < .001$). Results of median splits are displayed in Supplementary Material 6.1.

Additional files

Supplementary material [↗](#)

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Author information

Samantha K Millard

Faculty of Medicine, University of New South Wales (UNSW), Sydney, Australia, Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia
ORCID iD: [0000-0003-1409-8179](https://orcid.org/0000-0003-1409-8179)

For correspondence: s.millard@neura.edu.au

Alan KI Chiang

Faculty of Medicine, University of New South Wales (UNSW), Sydney, Australia, Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia
ORCID iD: [0000-0002-9156-0534](https://orcid.org/0000-0002-9156-0534)

Peter Humburg

Mark Wainwright Analytical Centre, University of New South Wales (UNSW), Sydney, Australia
ORCID iD: [0000-0002-3331-6496](https://orcid.org/0000-0002-3331-6496)

Nahian Chowdhury

Faculty of Medicine, University of New South Wales (UNSW), Sydney, Australia, Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia
ORCID iD: [0000-0001-6357-0746](https://orcid.org/0000-0001-6357-0746)

Raafay Rehan

Faculty of Medicine, University of New South Wales (UNSW), Sydney, Australia, Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia

Andrew J Furman

Department of Radiology and Nuclear Medicine, University of Maryland School of Medicine, Baltimore, United States, Department of Neural and Pain Sciences, School of Dentistry, University of Maryland Baltimore, Baltimore, United States, Center to Advance Chronic Pain Research, University of Maryland Baltimore, Baltimore, United States
ORCID iD: [0000-0001-8055-9610](https://orcid.org/0000-0001-8055-9610)

Ali Mazaheri

School of Psychology, University of Birmingham, Birmingham, United Kingdom, Centre for Human Brain Health (CHBH), University of Birmingham, Birmingham, United Kingdom
ORCID iD: [0000-0001-5732-7555](https://orcid.org/0000-0001-5732-7555)

Siobhan M Schabrun

Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia, The Gray Centre for Mobility and Activity, Parkwood Institute, London, Canada, School of Physical Therapy, University of Western Ontario, London, Canada
ORCID iD: [0000-0002-9083-3107](https://orcid.org/0000-0002-9083-3107)

David A Seminowicz

Centre for Pain IMPACT, Neuroscience Research Australia (NeuRA), Sydney, Australia, Department of Neural and Pain Sciences, School of Dentistry, University of Maryland Baltimore, Baltimore, United States, Center to Advance Chronic Pain Research, University of Maryland Baltimore, Baltimore, United States, Department of Medical Biophysics, Schulich School of Medicine & Dentistry, University of Western Ontario, London, Canada
ORCID iD: [0000-0003-3111-3756](https://orcid.org/0000-0003-3111-3756)

Editors

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Reviewer #1 (Public review):

Summary:

In this study, Millard and colleagues investigated if the analgesic effect of nicotine on pain sensitivity, assessed with two pain models, is mediated by Peak Alpha Frequency (PAF) recorded with resting state EEG. The authors found indeed that nicotine (4 mg, gum) reduced pain ratings during phasic heat pain but not cuff pressor algometry compared to placebo conditions. Nicotine also increased PAF (globally). However, mediation analysis revealed that the reduction in pain ratings elicited by the phasic heat pain after taking nicotine was not mediated by the changes in PAF. Also, the authors only partially replicated the correlation between PAF and pain sensitivity at baseline (before nicotine treatment). At the group-level

no correlation was found, but an exploratory analysis showed that the negative correlation (lower PAF, higher pain sensitivity) was present in males but not in females. The authors discuss the lack of correlation.

In general, the study is rigorous, methodology is sound and the paper is well written. Results are compelling and sufficiently discussed.

Strengths:

Strengths of this study are the pre-registration, proper sample size calculation and data analysis. But also the presence of the analgesic effect of nicotine and the change in PAF.

Weaknesses:

It would even be more convincing if they had manipulated PAF directly.

<https://doi.org/10.7554/eLife.91933.2.sa3>

Reviewer #2 (Public review):

Summary:

The study by Millard et al. investigates the effect of nicotine on alpha peak frequency and pain in a very elaborate experimental design. According to the statistical analysis, the authors found a factor-corrected significant effect for prolonged heat pain but not for alpha peak frequency in response to the nicotine treatment.

Strengths:

I very much like the study design and that the authors followed their research line by aiming to provide a complete picture of the pain-related cortical impact of alpha peak frequency. This is very important work, even in the absence of any statistical significance. I also appreciate the preregistration of the study and the well-written and balanced introduction.

Weaknesses:

The weakness of the study revolves around two aspects:

(1) Source separation (ICA or similar) would have been more appropriate than electrode ROIs to extract the alpha signal. By using a source separation approach, different sources of alpha (mu, occipital alpha, laterality) could be disentangled.

(2) There is also a suggestion in the literature in the manuscript) that nicotine treatment may not work as intended. Instead, the authors' decision to use nicotine to modulate peak alpha frequency and pain was based on other, inappropriate work on chronic pain and chronic smokers. In the present study, the authors use nicotine treatment and transient painful stimulation in nonsmokers. The unfortunate decision to use nicotine severely hampered the authors' goal of the study.

Impact: The impact of the study could be to show what did not work to answer the authors' research questions. The study would have more impact with a more appropriate pain intervention model and an analysis strategy that untangles the different alpha sources.

<https://doi.org/10.7554/eLife.91933.2.sa2>

Reviewer #3 (Public review):

In this manuscript, Millard et al. investigate the effects of nicotine on pain sensitivity and peak alpha frequency (PAF) in resting state EEG. To this end, they ran a randomized, double-blind, placebo-controlled experiment involving 62 healthy adults that received either 4 mg nicotine gum (n=29) or placebo (n=33). Prolonged heat and pressure were used as pain models. Resting state EEG and pain intensity (assessed with a visual analog scale) were measured before and after the intervention. Additionally, several covariates (sex at birth, depression and anxiety symptoms, stress, sleep quality, among others) were recorded. Data was analyzed using ANCOVA-equivalent two-wave latent change score models, as well as repeated measures analysis of variance. Results do not show experimentally relevant changes of PAF or pain intensity scores for neither of the prolonged pain models due to nicotine intake.

The main strengths of the manuscript are its solid conceptual framework and the thorough experimental design. The researchers make a good case in the introduction and discussion for the need to further investigate the association of PAF and pain sensitivity. Furthermore, they proceed to carefully describe every aspect of the experiment in great detail, which is excellent for reproducibility purposes. Finally, they analyze the data from different and provide an extensive report of their results.

There are relevant weaknesses to highlight. Firstly, authors preregistered the study and the analysis plan, but the preregistration does not contain an estimation of the expected effect sizes or the rationale for the selected sample size. Furthermore, the authors interpret their results in a way that is not supported by the evidence (which is notorious in the abstract and the first paragraph of the discussion). Even though some of the differences are statistically significant (e.g., global PAF, pain intensity ratings during heat pain), these differences are far from being experimentally or clinically relevant. The effect sizes observed are not sufficiently large to consider that pain sensitivity was modulated by the nicotine intake, which puts into question all the answers to the research questions posed in the study. The authors attempt to nuance this throughout the discussion, but in a way that is not compatible with the main claims.

<https://doi.org/10.7554/eLife.91933.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:**Reviewer #1 (Public Review):***Summary:*

In this study, Millard and colleagues investigated if the analgesic effect of nicotine on pain sensitivity, assessed with two pain models, is mediated by Peak Alpha Frequency (PAF) recorded with resting state EEG. The authors found indeed that nicotine (4 mg, gum) reduced pain ratings during phasic heat pain but not cuff pressor algometry compared to placebo conditions. Nicotine also increased PAF (globally). However, mediation analysis revealed that the reduction in pain ratings elicited by the phasic heat pain after taking nicotine was not mediated by the changes in PAF. Also, the authors only partially replicated the correlation between PAF and pain sensitivity at baseline (before nicotine treatment). At the group-level no correlation was found, but an exploratory

analysis showed that the negative correlation (lower PAF, higher pain sensitivity) was present in males but not in females. The authors discuss the lack of correlation.

In general, the study is rigorous, methodology is sound and the paper is well-written. Results are compelling and sufficiently discussed.

Strengths:

Strengths of this study are the pre-registration, proper sample size calculation, and data analysis. But also the presence of the analgesic effect of nicotine and the change in PAF.

Weaknesses:

It would even be more convincing if they had manipulated PAF directly.

We thank Reviewer #1 for their positive and constructive comments regarding our study. We appreciate the view that the study was rigorous and methodologically sound, that the paper was well-written, and that the strengths included our pre-registration, sample size calculation, and data analysis.

In response to the reviewer's comment about more directly manipulating Peak Alpha Frequency (PAF), we agree that such an approach could provide a more direct investigation of the role of PAF in pain processing. We chose nicotine to modulate PAF as the literature suggested it was associated with a reliable increase in PAF speed. As mentioned in our Discussion, there are several alternative methods to manipulate PAF, such as non-invasive brain stimulation techniques (NIBS) like transcranial alternating current stimulation (tACS) or neurofeedback training. These approaches could help clarify whether a causal relationship exists between PAF and pain sensitivity. Although methods such as NIBS still require further investigation as there is little evidence for these approaches changing PAF (Millard et al., 2024).

Reviewer #2 (Public Review):

Summary:

The study by Millard et al. investigates the effect of nicotine on alpha peak frequency and pain in a very elaborate experimental design. According to the statistical analysis, the authors found a factor-corrected significant effect for prolonged heat pain but not for alpha peak frequency in response to the nicotine treatment.

Strengths:

I very much like the study design and that the authors followed their research line by aiming to provide a complete picture of the pain-related cortical impact of alpha peak frequency. This is very important work, even in the absence of any statistical significance. I also appreciate the preregistration of the study and the well-written and balanced introduction. However, it is important to give access to the preregistration beforehand.

Weaknesses:

The weakness of the study revolves around three aspects:

(1) I am not entirely convinced that the authors' analysis strategy provides a sufficient signal-to-noise ratio to estimate the peak alpha frequency in each participant reliably. A source separation (ICA or similar) would have been better suited than electrode ROIs to extract the alpha signal. By using a source separation approach, different sources of alpha (μ , occipital alpha, laterality) could be disentangled.

(2) Also, there's a hint in the literature (reference 49 in the manuscript) that the nicotine treatment may not work as intended. Instead, the authors' decision to use nicotine to modulate the peak alpha frequency and pain relied on other, not suitable work on chronic pain and permanent smokers. In the present study, the authors use nicotine treatment and transient painful stimulation on nonsmokers.

(3) In my view, the discussion could be more critical for some aspects and the authors speculate towards directions their findings can not provide any evidence. Speculations are indeed very important to generate new ideas but should be restricted to the context of the study (experimental pain, acute interventions). The unfortunate decision to use nicotine severely hampered the authors' aim of the study.

Impact:

The impact of the study could be to show what has not worked to answer the research questions of the authors. The authors claim that their approach could be used to define a biomarker of pain. This is highly desirable but requires refined methods and, in order to make the tool really applicable, more accurate approaches at subject level.

We thank reviewer #2 for their recognition of the study's design, the importance of this research area, and the pre-registration of our study. In response to the weaknesses highlighted:

(1) We appreciate the reviewer's suggestion to improve the signal-to-noise ratio by applying source separation techniques, such as ICA, which have now been performed and incorporated into the manuscript. Our original decision to use sensor-level ROIs followed the precedent set in previous studies, our rationale being to improve reproducibility and avoid biases from picking individual electrodes or manually picking sources. We have added analyses using an automated pipeline that selects components based on the presence of a peak in the alpha range and alignment with a predefined template topography representing sensorimotor sites. Here again we found no significant differences in the mediation results that used a sensor space sensorimotor ROI, further supporting the robustness of the chosen approach. ICA could still potentially disentangle different sources of alpha, such as occipital alpha and mu rhythm, and provide new insights into the PAF-pain relationship. We have now added a discussion in the manuscript about the potential advantages of source separation techniques and suggest that the possible contributions of separate alpha sources be investigated and compared to sensor space PAF as a direction for future research.

(2) We recognise the reviewer's concern regarding our choice of nicotine as a modulator of pain and alpha peak frequency (PAF). The meta-analysis by Ditte et al. (2016) indeed points to small effect sizes for nicotine's impact on experimental pain and highlights the potential for publication bias. However, our decision to use nicotine in this study was not primarily based on its direct analgesic effects, but rather on its well-documented ability to modulate PAF, in smoking and non-smoker populations, as outlined in our study aims.

In this regard, the intentional use of nicotine was to assess whether changes in PAF could mediate alterations in pain. This approach aligns with the broader concept that a direct effect of an intervention is not necessary to observe indirect effects (Fairchild & McDaniel, 2017). We have, however, revised our introduction to further clarify this rationale, highlighting that nicotine was used as a tool for PAF modulation, not solely for its potential analgesic properties.

(3) We agree with the reviewer's observation that certain aspects of the Discussion could be more cautious, particularly regarding speculations about nicotine's effects and PAF as a biomarker of pain. We have revised the Discussion to ensure that our interpretations are better grounded in the data from this study, clearly stating the limitations and avoiding

overgeneralization. This revision focuses on a more critical evaluation of the potential relationships between PAF, nicotine, and pain sensitivity based solely on our experimental context.

Finally, We also apologize for not providing access to the preregistration earlier. This was an oversight on our end, and we will ensure that future preregistrations are made available upfront.

Reviewer #3 (Public Review):

*In this manuscript, Millard et al. investigate the effects of nicotine on pain sensitivity and peak alpha frequency (PAF) in resting state EEG. To this end, they ran a pre-registered, randomized, double-blind, placebo-controlled experiment involving 62 healthy adults who received either 4 mg nicotine gum (n=29) or placebo (n=33). Prolonged heat and pressure were used as pain models. Resting state EEG and pain intensity (assessed with a visual analog scale) were measured before and after the intervention. Additionally, several covariates (sex at birth, depression and anxiety symptoms, stress, sleep quality, among others) were recorded. Data was analyzed using ANCOVAequivalent two-wave latent change score models, as well as repeated measures analysis of variance. Results do not show *experimentally relevant* changes of PAF or pain intensity scores for either of the prolonged pain models due to nicotine intake.*

The main strengths of the manuscript are its solid conceptual framework and the thorough experimental design. The researchers make a good case in the introduction and discussion for the need to further investigate the association of PAF and pain sensitivity. Furthermore, they proceed to carefully describe every aspect of the experiment in great detail, which is excellent for reproducibility purposes. Finally, they analyse the data from almost every possible angle and provide an extensive report of their results.

The main weakness of the manuscript is the interpretation of these results. Even though some of the differences are statistically significant (e.g., global PAF, pain intensity ratings during heat pain), these differences are far from being experimentally or clinically relevant. The effect sizes observed are not sufficiently large to consider that pain sensitivity was modulated by the nicotine intake, which puts into question all the answers to the research questions posed in the study.

We would like to express our gratitude to Reviewer #3 for their thoughtful and constructive review, including the positive feedback on the strengths of our study's conceptual framework, experimental design, and thorough methodological descriptions.

We acknowledge the concern regarding the experimental and clinical relevance of some statistically significant results (e.g., global PAF and pain intensity during heat pain) and agree that small effect sizes may limit their practical implications. However, our primary goal was to assess whether nicotine-induced changes in PAF mediate pain changes, rather than to demonstrate large direct effects on pain sensitivity. Nicotine was chosen for its known ability to modulate PAF, and our focus was on the mechanistic role of PAF in pain perception. To clarify this, we have revised the discussion to better differentiate between statistical significance, experimental relevance, and clinical applicability. We emphasize that this study represents a preliminary step towards understanding PAF's mechanistic role in pain, rather than a direct clinical application.

We appreciate the suggestion to refine our interpretation. We have adjusted our language to ensure it aligns with the effect sizes observed and made recommendations for future research, such as testing different nicotine doses, to potentially uncover stronger or more clinically relevant effects.

Although modest, we believe these findings offer valuable insights into the potential mechanisms by which nicotine affects alpha oscillations and pain. We have also discussed how these small effects could become more pronounced in different populations (e.g., chronic pain patients) and over time, offering guidance for future research on PAF modulation and pain sensitivity.

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

I have a number of points that the authors may want to consider for this or future work.

(1) By reviewing the literature provided by the authors in the introduction I think that using nicotine as a means to modulate pain and alpha peak frequency was a mistake. The only work that may give a hint on whether nicotine can modulate experimental pain is the meta-analysis by Ditre and colleagues (2016). They suggest that their small effect may contain a publication bias. I think the other "large body of evidence" is testing something else than analgesia.

Thank you for your consideration of our choice of nicotine in the study. The meta-analysis by Ditre and colleagues (2016) suggests small effect sizes for nicotine's impact on experimental pain, compared to the moderate effects claimed in some papers, especially when accounting for the potential publication bias you mentioned. However, our selection of nicotine was primarily driven by its documented ability to modulate PAF rather than its direct analgesic effects, as clearly stated in our aims. Therefore, we do not view our decision to use nicotine as a mistake; instead, it was aligned with our goal of assessing whether changes in PAF mediate alterations in pain and thus served as a valuable tool. This perspective aligns with the broader concept that a direct effect is not a prerequisite for observing indirect effects of an intervention on an outcome (Fairchild &

McDaniel, 2017). To further enhance clarity, we've revised the introduction to emphasize the role of nicotine in manipulating PAF in relation to our study's aims.

Previously we wrote: "A large body of evidence suggests that nicotine is an ideal choice for manipulating PAF, as both nicotine and smoking increase PAF speed [37,40–47] as well as pain thresholds and tolerance [48–52]." This has been changed to read: "Because evidence suggests that nicotine can modulate PAF, where both nicotine and smoking increase PAF speed [37,40–47], we chose nicotine to assess our aim of whether changes in PAF mediate changes in pain in a 'mediation by design' approach [48]. In addition, given evidence that nicotine may increase experimental pain thresholds and tolerance [49–53], nicotine could also influence pain ratings during tonic pain."

(2) As mentioned above, the OSF page is not accessible.

We apologise for this. We had not realised that the pre-registration was under embargo, but we have now made it available.

(3) I generally struggle with the authors' approach to investigating alpha. With the approach the authors used to detect peak alpha frequency it might be that the alpha signal may just show such a low amplitude that it is impossible to reliably detect it at electrode level. In my view, the approach is not accurate enough, which can be seen by the "jagged" shape of the individual alpha peak frequency. In my view, a source separation technique would have been more useful. I wonder which of the known cortical alphas contributes to the effects the authors have reported previously: occipital, mu rhythms projections or something else? A source separation approach disentangles the

different alphas and will increase the SNR. My suggestion would be to work on ICA components or similar approaches. The advantage is that the components are almost completely free of any artefacts. ICAs could be run on the entire data or separately for each individual. In the latter case, it might be that some participants do not exhibit any alpha component.

We appreciate your thoughtful consideration of our approach to investigating alpha. The calculation of PAF involves various methods and analysis steps across the literature (Corcoran et al., 2018; Gil Avila et al., 2023; McLain et al., 2022). Your query about which known cortical alphas contribute to reported effects is important. Initially focusing on a sensorimotor component from an ICA in Furman et al., 2018, subsequent work from our labs suggested a broader relationship between PAF and pain across the scalp (Furman et al., 2019; Furman et al., 2020; Millard et al., 2022), and a desire to conduct analyses at the sensor level in order to improve the reproducibility of the methods (Furman et al., 2020). However, based on your comment we have made several additions to the manuscript, including: explaining why we did not use manual ICA methods, suggest this for future research, and added an exploratory analysis using a recently developed automated pipeline that selects components based on the presence of a peak in the alpha range and alignment with a predefined template topography representing activity from occipital or motor sites.

While we acknowledge that ICA components can offer a better signal-to-noise ratio (SNR) and possibly smoother spectral plots, we opted for our chosen method to avoid potential bias inherent in deciding on a component following source separation. The desire for a quick, automated, replicable, and unbiased pipeline, crucial for potential clinical applications of PAF as a biomarker, influenced this decision. At the time of analysis registration, automated methods for deciding which alpha components to extract following ICA were not apparent. We have now added this reasoning to Methods.

“Contrary to some previous studies that used ICA to isolate sensory region alpha sources (Furman et al., 2018; De Martino et al., 2021; Valentini et al., 2022), we used pre-determined sensor level ROIs to improve reproducibility and reduce the potential for bias when individually selecting ICA components. Using sensor level ROIs may decrease the signal-to-noise ratio of the data; however, this approach has still been effective for observing the relationship between PAF and experimental pain (Furman et al., 2019; Furman et al., 2020).”

We have also added use of ICA and development of methods as a suggestion for future research in the discussion:

“Additionally, the use of global PAF may have introduced mediation measurement error into our mediation analysis. The spatial precision used in the current study was based on previous literature on PAF as a biomarker of pain sensitivity, which have used global and/or sensorimotor ROIs (Furman et al., 2018; Furman et al., 2020). Identification and use of the exploratory electrode clusters found in this study could build upon the current work (e.g., Furman et al., 2021). However, exploratory analysis of the clusters found in the present analysis demonstrated no influence on mediation analysis results (Supplementary Materials 3.8-3.10). Alternatively, independent component analysis (ICA) could be used to identify separate sources of alpha oscillations (Choi et al., 2005), as used in other experimental PAF-pain studies (Furman et al., 2018; Valentini et al., 2022), which could aid to disentangle the potential relevance of different alpha sources in the PAFpain relationship. Although this comes with the need to develop more reproducible and automated methods for identifying such components.”

The specific location or source of PAF that relates to pain remains unclear. Because of this, we did employ an exploratory cluster-based permutation analysis to assess the potential for variations in the presence of PAF changes across the scalp at sensor level, and emphasise that location of PAF change could be explored in future. However, we have now conducted the

mediation analysis (difference score 2W-LCS model) using averages from the data-driven parietal cluster, frontal cluster, and both clusters together. For these we see a stronger effect of gum on PAF change, which was expected given the data driven approach of picking electrodes. There was still a total and direct effect of nicotine on pain during the PHP model, but still no indirect effect via change in PAF. For the CPA models, there were still no significant total, direct, or indirect effects of nicotine on CPA ratings. Therefore, using these data-driven clusters did not alter results compared to the model using the global PAF variable.

The reader has been directed to this supplementary material so:

“The potential mediating effect of this change in PAF on change in PHP and CPA was explored (not pre-registered) by averaging within each cluster (central-parietal: CP1, CP2, Cpz, P1, P2, P3, P4, Pz, POz; right-frontal: F8, FT8, FT10) and across both clusters. This averaging across electrodes produced three new variables, each assessed in relation to mediating effects on PHP and CPA ratings. The resulting in six exploratory mediation analysis (difference score 2W-LCS) models demonstrated minimal differences from the main analysis of global PAF (8-12 Hz), except for the

expected stronger effect of nicotine on change in PAF ($bs = 0.11-0.14$, $ps < .003$; Supplementary Materials 3.8-3.10).”

Moreover, our team has been working on an automated method for selecting ICA components, so in response to your comment we assessed whether using this method altered the results of the current analysis. The in-depth methodology behind this new automatic pipeline will be published with a validation from some co-authors in the current collaboration in due course. At present, in summary, this automatic pipeline conducts independent component analysis (ICA) 10 times for each resting state, and selects the component with the highest topographical correlation to a template created of a sensorimotor alpha component from Furman et al., (2018).

The results of the PHP or CPA mediation models were not substantially different using the PAF calculated from independent components than that using the global PAF. For the PHP model, the total effect ($b = -0.648$, $p = .033$) and direct effects ($b = -0.666$, $p = .035$) were still significant, and there was still no significant indirect effect ($b = 0.018$, $p = .726$). The general fit was reduced, as although the CFI was above 0.90, akin to the original model, the RMSEA and SRMR were not below 0.08, unlike the original models (Little, 2013). For the CPA model, there were still no significant total ($b = -0.371$, $p = .357$), direct ($b = -0.364$, $p = .386$), or indirect effects ($b = -0.007$, $p = .906$), and the model fit also decreased, with CFI below 0.90 and RMSEA and SRMR above 0.08. See supplementary material (3.11). Note that still no correlations were seen between this IC sensorimotor PAF and pain (PHP: $r = 0.11$, $p = .4$; CPA: $r = -0.064$, $p = .63$).

Interestingly, in both models, there was now no longer a significant a-path (PHP: $b = 0.08$, $p = 0.292$; CPA: $b = 0.039$, $p = 0.575$), unlike previously observed (PHP: $b = 0.085$, $p = 0.018$; CPA: $b = 0.089$, $p = 0.011$). We interpret this as supporting the previously highlighted difference between finding an effect on PAF globally but not in a sensorimotor ROI (and now a sensorimotor IC), justifying the exploratory CBPA and the suggestion in the discussion to explore methodology.

We understand that this analysis does not fully uncover the reviewer’s question in which they wondered which of the known cortical alphas contributes to the effects reported in our previous work. However, we consider this exploration to be beyond the scope of the current paper, as it would be more appropriately addressed with larger datasets or combinations of datasets, potentially incorporating MEG to better disentangle oscillatory sources. The highlighted differences seen between global PAF, sensorimotor ROI PAF, sensorimotor IC PAF,

as well as the CBPA of PAF changes provide ample directions for future research to build upon: 1) which alpha (sensor or source space) are related to pain, 2) how are these alpha signals represented robustly in a replicable way, and 3) which alpha (sensor or source space) are manipulable through interventions. These are all excellent questions for future studies to investigate.

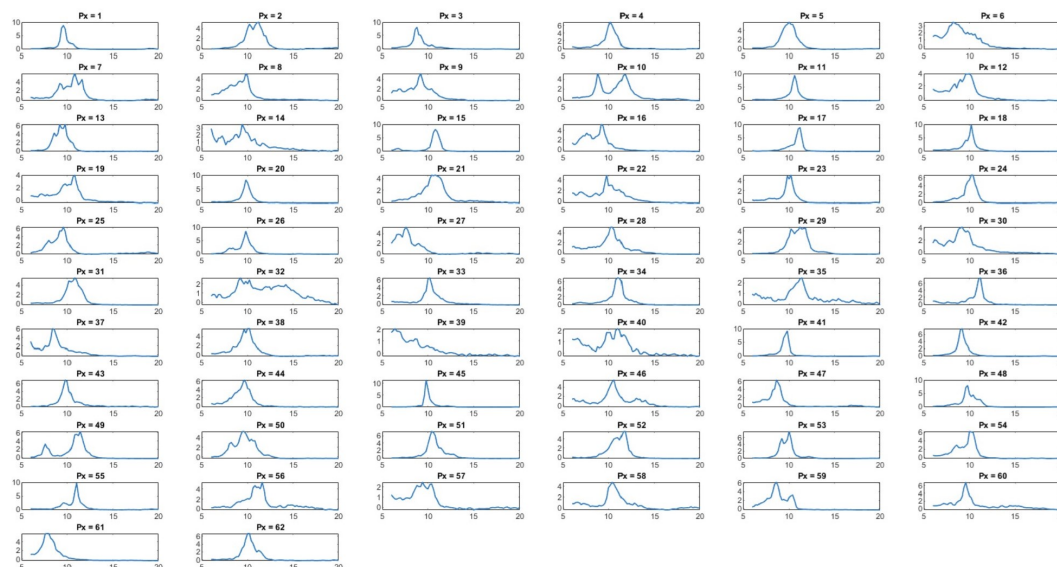
The below text has been added to the Discussion:

In-house code was developed to compare a sensorimotor component to the results presented in this manuscript (Supplementary Material 3.11), showing similar results to the sensorimotor ROI mediation analysis presented here. However, examination of which alpha - be it sensor or source space - are related to pain, how they can be robustly represented, and how they can be manipulated are ripe avenues for future study.

(4) I have my doubts that you can get a reliable close to bell-shaped amplitude distribution for every participant. The argument that the peak detection procedure is hampered by the high-amplitude lower frequency can be easily solved by subtracting the "slope" before determining the peak. My issue is that the entire analysis is resting on the assumption that each participant has a reliable alpha effect at electrode level. This is not the case. Non-alpha participants can severely distort the statistics. ICA-based analyses would be more sensitive but not every participant will show alpha. You may want to argue with robust group effects but In my view, every single participant counts, particularly for this type of data analysis, where in the case of a low SNR the "peak" can easily shift to the extremes. In case there is an alpha effect for a specific subject, we should see a smooth bump in the frequency spectrum between 8 and 12 Hz. Anything beyond that is hard to believe. The long stimulation period allows a broad FFT analysis window with a good frequency resolution in order to detect the alpha frequency bump.

The reviewer is correct that non-alpha participants can distort the statistics. We did visually assess the EEG of each individual's spectra at baseline to establish the presence of global peaks, as we believe this is good practice to aid understanding of the data. Please see Author response image 1 for individual spectra seen at baseline. Although not all participants had a 'smooth bump in the frequency spectrum between 8 and 12 Hz', we prefer to not apply/necessitate this assumption to our data. Chiang et al., (2011) suggest that ~3% of individuals do not have a discernible alpha peak, and in our data we observed only one participant without a very obvious spectral peak (px-39). But, this participant does have enough activity within the alpha range to identify PAF by the CoG method (i.e. not just flat spectra and activity on top of 1/f characteristics). Without a pre-registered and standardised decision process to remove such a participant in place, we opted to not remove any participants to avoid curation of our data.

Author response image 1.



(5) I find reports on frequent channel rejections reflect badly on the data quality. Bad channels can be avoided with proper EEG preparation. EEG should be continuously monitored during recording in order to obtain best data quality. Have any of the ROI channels been rejected?

We appreciate your attention to the channel rejection. We believe that the average channels removed (0.94, 0.98, 0.74, and 0.87 [range: 0-4] for each of the four resting states out of 64 channels) does not suggest overly frequent rejection, as it was less than one electrode on average and the numbers are below the accepted number of bad channels to remove/interpolate (i.e. 10%) in EEG pipelines (Debnath et al., 2020; Kayhan et al., 2022). To maintain data quality, consistently poor channels were identified and replaced over time. We hope you will accept our transparency on this issue and note that by stating how channel removal decisions were made (i.e. 8 or more deviations) and reporting the number of channels removed, we adhere to the COBIDAS guidelines (Pernet et al., 2018; 2020).

During analysis, cases of sensorimotor ROI channels being rejected were noted and are now specified in our manuscript. “Out of 248 resting states recorded, 14 resting states had 4 ROI channels instead of 5. Importantly, no resting state had fewer than 4 channels for the sensorimotor ROI.”

Note, we also realised that we had not specified that we did interpolate channels for the cluster based permutation analysis. This has been corrected with the following sentence:

“Removed channels were not interpolated for the pre-registered global and sensorimotor ROI averaged analyses, but were interpolated for an exploratory cluster based permutation analysis using the nearest neighbour average method in Fieldtrip.”

(6) I have some issues buying the authors' claims that there is an effect of nicotine on prolonged pain. By looking at the mean results for the nicotine and placebo condition, this can not be right. What was the point in including the variables in the equation? In my view, in this within-subject design the effect of nicotine should be universal, no matter what gender, age, or depression. The unconditional effect of nicotine is close to zero. I

can not get my head around how any of the variables can turn the effects into significance. There must be higher or lower variable scores that might be related to a higher or lower effect on nicotine. The question is not to consider these variables as a nuisance but to show how they modulate the pain-related effect of nicotine treatment. Still, the overall nicotine effect of the entire group is basically zero.

Another point is that for within-subject analyses even tiny effects can become statistically significant if they are systematically in one direction. This might be the case here. There might be a significant effect of nicotine on pain but the actual effect size (5.73 vs. 5.78) is actually not interpretable. I think it would be interesting for the reader how (in terms of pain rating difference) each of the variables can change the effect of nicotine.

Thank you for your comments. We recognize the concern about interpreting the effect of nicotine on prolonged pain solely based on mean results, and in fact wish to discourage this approach. It's crucial to note that both PAF and pain are highly individual measures (i.e. high inter-individual variance), necessitating the use of random intercepts for participants in our analyses to acknowledge the inherent variability at baseline across participants. Including random intercepts rather than only considering the means helps address the heterogeneity in baseline levels among participants. We also recognise that displaying the mean PHP ratings for all participants in Table 2 could be misleading, firstly because these means do not have weight in an analysis that takes into account a random-effects intercept for participants, and secondly because two participants (one from each group) did not have post-gum PHP assessments and were not included in the mediation analysis due to list-wise deletion of missing data. Therefore, to reduce the potential for misinterpretation, we have added extra detail to display both the full sample and CPA mediation analysis (i.e. N=62) and the data used for PHP mediation analysis (i.e. n=60) in Table 2. We hope that the extra details added to this table will help the readers interpretation of results.

In light of this, we have also altered the PAF Table 3 to reflect both the pre-post values used for the CPA mediation and baseline correlations with CPA and PHP pain (i.e. N=62), and the pre-post values used for the PHP mediation (i.e. n=60).

It is inherently difficult to visualise the findings of a mediation analysis with confounding variables that also used latent change scores (LCS) and random-effect intercepts for participants. LCS was specifically used because of issues of regression to the mean that occur if you calculate a straightforward 'difference-score', therefore calculating the difference in order to demonstrate the results of the statistical model in a figure, for example, does not provide a full description of the data assessed (Valente & McKinnon, 2017). Nevertheless, if we look at the data descriptively with this in mind, then calculating the change in PHP ratings does indicate that, for the nicotine group, the mean change in PHP ratings was -0.047 (SD = 1.05, range: -4.13, 1.45). Meanwhile, for the placebo group the mean change in PHP ratings was 0.33 (SD = 0.75, range: -1.37, 1.66). Therefore suggesting a slight decrease in pain ratings on average for the nicotine group compared to a slight increase on average for the placebo group. With control for pre-determined confounders, we found that the latent change score was -0.63 lower for the nicotine group compared to the control group (i.e. the direct effect of nicotine on change in pain).

If the reviewer is only discussing the effect of nicotine on pain, we do not believe that this effect 'should be universal'. There is clear evidence that effects of nicotine on other measures can vary greatly across individuals (Ettinger et al., 2009; Falco & Bevins, 2015; Pomerleau et al., 1995). Our intention would not be to propose a universal effect but to understand how these variables may influence nicotine's impact on pain for individuals. Here we focus on the effects of nicotine on PAF and pain sensitivity, but attempted to control for the potential influence of these other confounding factors. Therefore, our statistical approach goes beyond mean values, incorporating variables like sex at birth, age, and depression to control for and

explore potential modulating factors. Control for confounding factors is an important aspect of mediation analysis (Lederer et al., 2019; VanderWeele, 2019).

Regarding the seemingly small effect size, we understand your concern. Indeed ‘tiny effects can become statistically significant if they are systematically in one direction’, which may be what we see in this analysis. We do not agree that the effect is ‘not interpretable’, rather that it should be interpreted in light of its small effect size (effect size being the beta coefficient in our analysis, rather than the mean group difference). We agree on the importance of considering practical significance alongside statistical significance and hope to conduct additional experiments and analyses in future to elucidate the contribution of each variable to the subtle and therefore not entirely conclusive overall effect you mention.

Your feedback on this is valuable, and we have ensured a more detailed discussion in the revised manuscript on how these factors should be interpreted alongside some additional post-hoc analyses of confounding factors that were significant in our mediation, with the note that investigation of these interactions is exploratory. We had already discussed the potential contribution of sex on the effect of nicotine on PAF, with exploratory post-hoc analysis on this included in supplementary materials. In addition, we have now added an exploratory post-hoc analysis on the potential contribution of stress on the effect of nicotine on pain. This then shows the stratified effects by the covariates that our model suggest are influencing change in PAF and pain.

Results edits:

“There was also a significant effect of perceived stress at baseline on change in PHP ratings when controlling for group allocation and other confounding variables ($b = -0.096$, $p = .048$, bootstrapped 95% CI: $[-0.19, -0.000047]$), where higher perceived stress resulted in larger decreases in PHP ratings (see Supplementary Material 3.3 for post-hoc analysis of stress).”

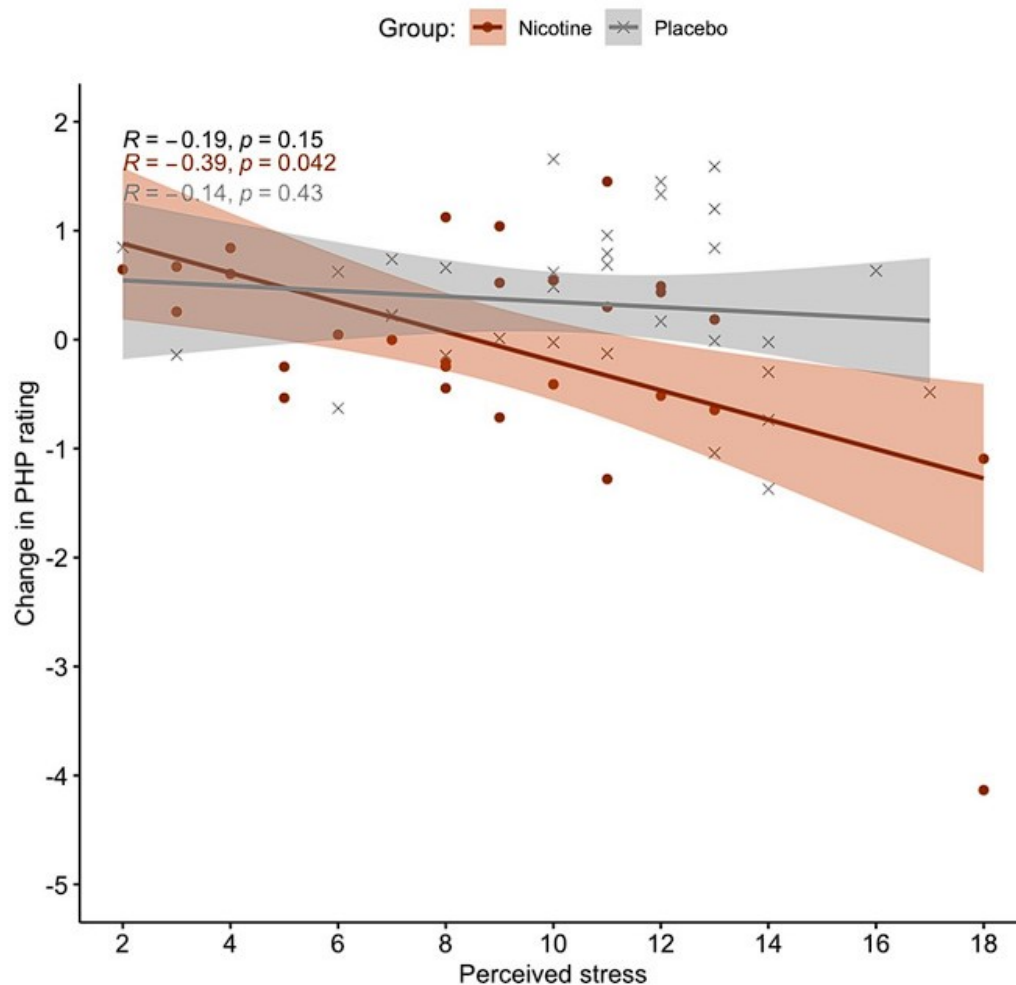
Supplementary material addition:

“3.3 Exploratory analysis of the influence of perceived stress on the effects of nicotine on change in PHP ratings “

“Due to the significant estimated effects of perceived stress on change in PHP ratings in the 2WLCS mediation model, we also explored post-hoc effects of stress on change in PHP ratings. We found that there is strong evidence for a negative correlation between stress and change in PHP rating within the nicotine group ($n = 28$, $r = -0.39$, $BF_{10} = 13.65$; Figure 3) that is not present in the placebo group, with equivocal evidence ($n = 32$, $r = -0.14$, $BF_{10} = 0.46$). This suggests that those with higher baseline stress who had nicotine gum experienced greater decreases in PHP ratings. Note that there was less, but still sufficient evidence for this relationship within the nicotine group when the participant who was a potential outlier for change in PHP rating was removed ($n = 27$, $r = -0.32$, $BF_{10} = 1.45$). “

Author response image 2.

Spearman correlations of baseline perceived stress with the change in phasic heat pain (PHP) ratings, suggest strong evidence for a negative relationship for the nicotine gum group in orange ($n=28$; $BF_{10}=13.65$) but not for the placebo group in grey ($n=32$; $BF_{10}=0.46$). Regression lines and 95% confidence intervals.



Discussion edits:

“For example, in addition to the effect of nicotine on prolonged heat pain ratings, our results suggest an effect of stress on changes in heat pain ratings, with those self-reporting higher stress at baseline having greater reductions in pain. Our post-hoc analysis suggested that this relationship between higher stress and larger decrease in PHP ratings was only present for the nicotine group (Supplementary Material 3.3). As stress is linked to nicotine use [69,70] and pain [71–73], these interactions should be explored in future.”

(7) Is the differential effect of nicotine vs. placebo based on the pre vs. post treatment effect of the placebo condition or on the pre vs. post effect of the nicotine treatment? Can the mediation model be adapted and run for each condition separately? The placebo condition seems to have a stronger effect and may have driven the result.

Thank you for your comments. In our mediation analysis, the differential effect of nicotine vs. placebo is assessed as a comparison between the pre-post difference within each condition. A latent change score (i.e. pre-post) is calculated for each condition (nicotine and placebo), and then the effect of being in the nicotine group (dummy coded as 1) is compared to being in the placebo group (dummy coded as 0). The comparison between conditions is needed for this model (Valente & MacKinnon, 2017), as we are assessing the change in PAF and pain in the nicotine group compared to the change in the placebo group.

However, to address your response, it is possible to simplify and assess the relationship between the change in peak alpha frequency (PAF) and change in pain within each gum group (nicotine and placebo) independently, without including the intervention as a factor. To do this, the mediation model can be simplified to regression analysis with latent change scores that focus purely on these relationships. The results of this can help to understand whether change in PAF influences change in pain within each group separately. As with the main analysis, we see no significant influence of change in PAF on change in pain while controlling for the same confounding variables within the nicotine group (Beta = -0.146 +/- 1.105, $p = 0.895$, 95% CI: -2.243, 2.429) or the placebo group (Beta = 0.730 +/- 2.061, $p = 0.723$, 95% CI: -4.177, 3.625).

When suggesting that the “the placebo condition seems to have a stronger effect and may have driven the result”, we believe you are referring to the increase in mean PHP ratings within the placebo group from pre (5.51 +/- 2.53) to post-placebo gum (5.84 +/- 2.67). Indeed there was a significant increase in pain ratings pre to post chewing placebo gum ($t(31) = -2.53$, $p = 0.0165$, 95% CI: -0.603, -0.0653), that was not seen after chewing nicotine gum ($t(27) = 0.237$, $p = 0.81$, 95% CI: -0.358, 0.452). In lieu of a control where no gum was chewed (i.e. simply a second pain assessment ~30 minutes after the first), we assume the gum without nicotine is a good reference that controls for the effect of time plus expectation of chewing nicotine gum. With this in mind, as we describe in our results, the change in PHP ratings is reduced in the nicotine group compared to the placebo group. Note that this phrasing keeps the effect of placebo on pain as our reference from which to view the effect of nicotine on pain. However, you are correct that we need to ensure we emphasise that the change in pain in the PHP group is reduced in comparison to the change seen after placebo.

We have not included these extra statistics in our revised manuscript, but hope that they aid the your understanding and interpretation of the included analyses and have highlighted these nuances in the discussion.

“However, we note that the observed effect of nicotine on pain was small in magnitude, and most prominent in comparison to the effect of placebo, where pain ratings increased after chewing, which brings into question whether this reduction in pain is meaningful in practice.”

(8) I would not dare to state that nicotine can function as an acute analgesic. Acute analgesics need to work for everyone. The average effect here is close to zero.

In light of your feedback, we have refined our language to avoid a sweeping assertion of universal analgesic effects and emphasize individual variability. Nicotine's role as a coping strategy for pain is acknowledged in the literature (Robinson et al., 2022), with the meta-analysis by Ditre et al. (2016) discussing its potential as an acute analgesic in humans, along with some evidence from animal research (Zhang et al., 2020). Our revised discussion underscores the need for further exploration into factors influencing nicotine's potential impact on pain. We have also specified the short-term nature of nicotine use in this context to distinguish acute effects from potential opposing effects after long-term use (Zhang et al., 2020).

“Short-term nicotine use is thought to have acute analgesic properties in experimental settings, with a review reporting that nicotine increased pain thresholds and pain tolerance [49]. In addition, research in a rat model suggests analgesic effects on mechanical thresholds after short-term nicotine use (Zhang et al., 2020). However, previous research has not assessed the acute effects of nicotine on prolonged experimental pain models. The present study found that 4 mg of nicotine reduced heat pain ratings during prolonged heat pain compared to placebo for our human participants, but that prolonged pressure pain decreased irrespective of which gum was chewed. Our findings are thus partly consistent with the idea

that nicotine may have acute analgesic properties [49], although further research is required to explore factors that may influence nicotine's potential impact on a variety of prolonged pain models. We further advance the literature by reporting this effect in a

model of prolonged heat pain, which better approximates the experience of clinical pain than short lasting models used to assess thresholds and tolerance [50]. However, we note that the observed effect of nicotine on pain was small in magnitude, and most prominent in comparison to the effect of placebo, where pain ratings increased after chewing, which brings into question whether this reduction in pain is meaningful in practice. Future research should examine whether effects on pain increase in magnitude with different nicotine administration regimens (i.e. dose and frequency)."

(9) Figures 2E and 2F are not particularly intuitive. Usually, the colour green in "jet" colour coding is being used for "zero" values. I would suggest to cut off the blue and use only the range between red green and red.

We have chosen to retain the current colour scale for several reasons. In our analysis, green represents the middle of the frequency range (approx 10 Hz in this case), and if we were to use green as zero, it would effectively remove both blue and green from the plot, resulting in only red shades. Additionally, we have provided a clear colour scale for reference next to the plot, which allows readers to interpret the data accurately. Our intention is to maintain clarity and precision in representing the data, rather than conforming strictly to conventional practices in color coding.

We believe that the current representation effectively conveys the results of our study while allowing readers to interpret the data within the context provided. Thank you again for your suggestion, and we hope you understand our reasoning in this matter.

(10) Did the authors do their analysis on the parietal ROI or on the pre-registered ROI?

The analysis was conducted on the pre-registered sensorimotor ROI and on the global values. We have now also conducted the analysis with the regions suggested with the cluster based permutation analysis as requested by reviewer 2, comment 3.

(11) Point 3.2 in the discussion. I would be very cautious to discuss smoking and chronic pain in the context of the manuscript. The authors can not provide any additional knowledge with their design targeting non-smokers, acute nicotine and experimental pain. The information might be interesting in the introduction in order to provide the reader with some context but is probably misleading in the discussion.

We appreciate your perspective and agree with your caution regarding the discussion of smoking and chronic pain. While our study specifically targets non-smokers and focuses on acute nicotine effects in experimental pain, we understand the importance of contextual clarity. We have removed these points from the discussion to not mislead the reader.

Previously we wrote, and have removed: "For those with chronic pain, smoking and nicotine use is reported as a coping strategy for pain [52]; abstinence can increase pain sensitivity [48,50], and pain is thus seen as a barrier to smoking cessation due to fear of worsening pain [51,52]. Therefore, continued understanding of the acute effects of nicotine on models of prolonged pain could improve understanding of the role of nicotine and smoking use in chronic pain [49,51,52]."

(12) I very much appreciate section 3.3 of the discussion. I would not give up on PAF as a target to modulate pain. A modulation might not be possible in such a short period of experimental intervention. PAF might need longer and different interventions to

gradually shift in order to attenuate the intensity of pain. As discussed by the authors themselves, I would also consider other targets for alpha analysis (as mentioned above not other electrodes or ROIs but separated sources.)

Thank you for your comments on section 3.3. We appreciate your recognition of the potential significance of PAF as a target for pain modulation. Your insights align with our considerations that the experimental intervention duration or type might be a limiting factor in observing substantial shifts in PAF to attenuate pain intensity. We had mentioned the use of the exploratory electrode clusters in future work, but have now also mentioned that the use of ICA to identify separate ICA sources may provide an alternative approach. See responses to your previous ICA comment regarding separate sources.

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Reviewer #3 (Recommendations For The Authors):

Introduction

(1) Rationale and link to chronic pain. I am not sure I agree with the statement "The ability to identify those at greater risk of developing chronic pain is limited". I believe there is an abundance of literature associating risk factors with the different instances of chronic pain (e.g., Mills et al., 2019). The fact that the authors cite studies involving potential neuroimaging biomarkers leads me to believe that they perhaps did not intend to make such a broad statement, or that they wanted to focus on individual prediction instead of population risk.

We thank the reviewer for the thought put into this comment. We did indeed wish to refer to individual prediction, but also realise that the focus on predicting pain might not be the most appropriate opening for this manuscript. Therefore, we have adjusted the below sentence to refer to the need to identify modifiable factors rather than the need to predict pain.

“Identifying modifiable factors that influence pain sensitivity could be a key step in reducing the presence and burden of chronic pain (van der Miesen et al., 2019; Davis et al., 2020; Tracey et al., 2021).”

(2) The statement "Individual peak alpha frequency (PAF) is an electro-physiological brain measure that shows promise as a biomarker of pain sensitivity, and thus may prove useful for predicting chronic pain development" is a non sequitur. PAF may very well be a biomarker of pain sensitivity, but the best measures of pain sensitivity we have (self-reported pain intensity ratings) in general are not in themselves predictive of the development of chronic pain. Conversely, features that are not related to pain sensitivity could be useful for predicting chronic pain (e.g., Tanguay-Sabourin et al., 2023).

We agree that it is essential to acknowledge that self-reported pain intensity ratings alone are not definitive predictors of chronic pain development. To align with this, we have revised the sentence, removing the second clause to avoid overstatement. The adjusted sentence now reads, "Individual peak alpha frequency (PAF) is an electrophysiological brain measure that shows promise as a biomarker of pain sensitivity."

(3) Finally, some of the statements in the discussion comparing a tonic heat pain model with chronic neuropathic pain might be an overstatement. Whereas it is true that some of the descriptors are similar, the time courses and mechanisms are vastly different.

We appreciate this comment, and agree that it is difficult to compare the heat pain model used to clinical neuropathic pain. This was an oversight and with further understanding we have removed this comment from the introduction and the discussion:

“In parallel, we saw no indication of a relationship between PAF and pain ratings during CPA. The introduction of the CPA model, specifically calibrated to a moderate pain threshold, provides further support for the notion that the relationship between PAF and pain is specific to certain pain types [17,28]. Prolonged heat pain was pre-dominantly described as moderate/severe shooting, sharp, and hot pain, whereas prolonged pressure pain was predominantly described as mild/moderate throbbing, cramping, and aching in the present study. It is possible that the PAF–pain relationship is specific to particular pain models and protocols [12,17].”

Methodology

(4) or the benefit of good science. However, I am compelled to highlight that I could not access the preregistered files, even though I waited for almost two weeks after requesting permission to do so. This was a problem on two levels: the main one is that I could not check the hypothesized effect sizes of the sample size estimation, which are not only central to my review, and in general negate all the benefits that should go with preregistration (i.e., avoiding phacking, publication bias, data dredging, HARKing, etc.). The second one is that I had to provide an email address to request access. This allows the authors to potentially identify the reviewers. Whereas I have no issues with this and I support transparent peer review practices (<https://elifesciences.org/inside-elifesciences/e3e90410/increasingtransparency-in-elifesciences-review-process>), I also note that this might condition other reviewers.

We apologise for this. We had not realised that the pre-registration was under embargo, but we have now made it available.

Interpretation of results

(5)To be perfectly clear, I trust the results of this study more than some of the cited studies regarding nicotine and pain because it was preregistered, the sample size is considerably larger, and it seems carefully controlled. I just do not agree with the interpretation of the results, stated in the first paragraph of the Discussion. Quoting J. Cohen, "The primary product of a research inquiry is one or more measures of effect size, not P values" (Cohen, 1990). As I am sure the authors are aware of, even tiny differences between conditions, treatments or groups will eventually be statistically significant given arbitrarily large sample sizes. What really matters then is the magnitude of these differences. In general, the authors hypothesize on why there were no differences on the pressure pain model, and why decreases in heat pain were not mediated by PAF, but do not seem to consider the possibility that the intervention just did not cause the intended effect on the nociceptive system, which would be a much more straightforward explanations for all observations.

While acknowledging and agreeing with the concern that 'even tiny differences between conditions, treatments, or groups will eventually be statistically significant given arbitrarily large sample sizes,' it's crucial to clarify that our sample size of N=62 does not fall into the category of arbitrarily large. We carefully considered the observed outcomes in the pressure pain model and the lack of PAF mediation in heat pain, as dictated by our statistical approach and the obtained results.

The suggestion of a straightforward explanation aligning with the intervention not causing the intended effect on the nociceptive system is a valid consideration. We did contemplate the

possibility of a false positive, emphasising this in the limitations of our findings and the need for replication to draw stronger conclusions to follow up this initial study.

*(6) In this regard, I do not believe that an average *increase* of 0.05 / 10 (Nicotine post - pre) can be considered a "reduction of pain ratings", regardless of the contrast with placebo (average increase of 0.24 / 10). This tiny effect size is more relevant in the context of the considerable inter-individual variation, in which subjects scored the same heat pain model anywhere from 1 to 10, and the same pressure pain model anywhere from 1 to 8.5. In this regard, the minimum clinically or experimentally important differences (MID) in pain ratings varies from study to study and across painful conditions but is rarely below 1 / 10 in a VAS or NRS scale, see f. ex. (Olsen et al., 2017). It is not my intention to question whether nicotine can function as an acute analgesic in general (as stated in the Discussion), but instead, if it worked as such under these very specific experimental conditions. I also acknowledge that the authors note this issue in two lines in the Discussion, but I believe that this is not weighed properly.*

We appreciate your perspective on the interpretation of the effect size, and we understand the importance of considering it in the context of individual variation.

As also discussed in response to comment 6 From reviewer 2, we recognize the concern about interpreting the effect of nicotine on prolonged pain solely based on mean results, and in fact wish to discourage this approach. It's crucial to note that both PAF and pain are highly individual measures (i.e. high inter-individual variance), necessitating the use of random intercepts for participants in our analyses to acknowledge the inherent variability at baseline across participants. Including random intercepts rather than only considering the means helps address the heterogeneity in baseline levels among participants. We also recognise that displaying the mean PHP ratings for all participants in Table 2 could be misleading, firstly because these means do not have weight in an analysis that takes into account a random-effects intercept for participants, and secondly because two participants (one from each group) did not have post-gum PHP assessments and were not included in the mediation analysis due to list-wise deletion of missing data. Therefore, to reduce the potential for misinterpretation, we have added extra detail to display both the full sample and CPA mediation analysis (i.e. N=62) and the data used for PHP mediation analysis (i.e. n=60) in Table 2. We hope that the extra details added to this table will help the readers interpretation of results.

Moreover, we have made sure refer to the comparison with the placebo group when discussing the reduction or decrease in pain seen in the nicotine group, for example:

"2) nicotine reduced prolonged heat pain intensity but not prolonged pressure pain intensity compared to placebo gum;"

"The nicotine group had a decrease in heat pain ratings compared to the placebo group and increased PAF speed across the scalp from pre to post-gum, driven by changes at central-parietal and right-frontal regions."

We have kept our original comment of whether this effect on pain is meaningful in practice to refer to the minimum clinically or experimentally important differences in pain ratings as highlighted by Olsen et al., 2017.

"While acknowledging the modest effect size, it's essential to consider the broader context of our study's focus. Assessing the clinical relevance of pain reduction is pertinent in applications involving the use of any intervention for pain management [69]. However, from a mechanistic standpoint, particularly in understanding the implications of and relation to PAF, the specific magnitude of the pain effect becomes less pivotal. Nevertheless, future

research should examine whether effects on pain increase in magnitude with different nicotine administration regimens (i.e. dose and frequency).”

(7) In line with the topic of effect sizes, average effect sizes for PAF in the study cited in the manuscript range from around 1 Hz (Boord et al., 2008; Wydenkeller et al., 2009; Lim et al., 2016), to 2 Hz (Foults et al., 1994), compared with changes of 0.06 Hz (Nicotine post - pre) or -0.01 Hz (Placebo post - pre). MIDs are not so clearly established for peak frequencies in EEG bands, but they should be certainly larger than some fractions of a Hertz (which is considerably below the reliability of the measurement).

We appreciate your care of these nuances. We acknowledge the differences in effect sizes between our study and those referenced in the manuscript. Given the current state of the literature, it's noteworthy that 'MIDs' for peak frequencies in EEG bands, particularly PAF changes, are not clearly established, other than a recent publication suggesting that even small changes in PAF are reliable and meaningful (Furman et al., 2021). In light of this, we have addressed the uncertainty around the existence and determination of MIDs in our revision, highlighting the need for further research in this area.

In addition, our study employed a greater frequency resolution (0.2 Hz) compared to some of the referenced studies, with approximately 0.5 Hz resolution (Boord et al., 2008; Wydenkeller et al., 2009; Foults et al., 1994). This improved resolution allows for a more precise measurement of changes in PAF. Considering this, it is plausible that studies with lower resolution might have conflated increases in PAF, and our higher resolution contributes to a more accurate representation of the observed changes.

We have also incorporated this insight into the manuscript, emphasising the methodological advancements in our study and their potential impact on the interpretation of PAF changes. Thank you for your thoughtful feedback.

“The ability to detect changes in PAF can be considerably impacted by the frequency resolution used during Fourier Transformations, an element that is overlooked in recent methodological studies on PAF calculation [16,95]. Changes in PAF within individuals might be obscured or conflated by lower frequency resolutions, which should be considered further in future research.”

(8) The authors also ran alternative statistical models to analyze the data and did not find consistent results in terms of PHP ratings (PAF modulation was still statistically significantly different). The authors attribute this to the necessity of controlling for covariates. Now, considering the effects sizes, aren't these statistically significant differences just artifacts stemming from the inclusion of too many covariates (Simmons et al., 2011)? How much influence should be attributable to depression and anxiety symptoms, stress, sleep quality and past pain, considering that these are healthy volunteers? Should these contrasting differences call the authors to question the robustness of the findings (i.e., whether the same data subjected to different analysis provides the same results), particularly when the results do not align with the preregistered hypothesis (PAF modulation should occur on sensorimotor ROIs)?

Thank you for your comments on our alternative statistical models. By including these covariates, we aim to provide a more nuanced understanding of the complexities within our data by considering their potential impact on the effects of interest. The decision to include covariates was preregistered (apologies again that this was not available) and made with consideration of balancing model complexity and avoiding potential confounding. Moreover, we hope that the insights gained from these analyses will offer valuable information about the behaviour of our data and aid future research in terms of power calculations, expected variance, and study design.

(9) Beyond that, I believe in some cases that the authors overreach in an attempt to provide explanations for their results. While I agree that sex might be a relevant covariate, I cannot say whether the authors are confirming a pre-registered hypothesis regarding the gender-specific correlation of PAF and pain, or if this is just a post hoc subgroup analysis. Given the large number of analyses performed (considering the main document and the supplementary files), caution should be exercised on the selective interpretation of those that align with the researchers' hypotheses.

We chose to explore the influence of sex on the correlation between PAF and pain, because this has also been investigated in previous publications of the relationship (Furman et al., 2020). We state that the assessment by sex is exploratory in our results on p.17: “in an exploratory analysis of separate correlations in males and females (Figure 5, plot C)”. For clarity regarding whether this was a pre-registered exploration or not, we have adjusted this to be: “in an exploratory analysis (not pre-registered) of separate correlations in males and females (Figure 5, plot C), akin to those conducted in previous research on this topic (Furman et al., 2020),

We have made sure to state this in the discussion also. Therefore, when we previously said on p.22:

“Regarding the relationship between PAF and pain at baseline, the negative correlation between PAF and pain seen in previous work [7–11,15] was only observed here for male participants during the PHP model for global PAF.” We have now changed this to: “Regarding the relationship between PAF and pain at baseline, the negative correlation between PAF and pain seen in previous work [7– 11,15] was only observed here for male participants during the PHP model for global PAF in an exploratory analysis.”

Please also note that we altered the colour and shape of points on the correlation plot (Figure 5 in initial submission), the male brown was changed to a dark brown as we realised that the light brown colour was difficult to read. The shape was then changed for male points so that the two groups can be distinguished in grey-scale.

Overall, your thoughtful feedback is instrumental in refining the interpretation of our findings, and we look forward to presenting a more comprehensive and nuanced discussion. Thank you for your comments.

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